Wake Parity and Highway Rigidity in Langton's Ant

Hao Ke
i@kehao.me

Abstract

What can the trail of a periodically translating Langton ant reveal about its turn sequence? We develop parity constraints on the odd-visit translation classes that form the persistent wake, and use geometric information to prove rigidity of the classical highway. For full drift $(2, -2)$, black growth twelve, transverse width ten and at most one odd class on each interior layer, the turn word is the standard period-104 word, up to cyclic phase. Parity restricts the profiles, an integer certificate bounds the period, and chronology-preserving enumeration completes the classification. With growth and interior thinness left free, every width-ten trace of full drift $(2, -2)$ satisfies the sharp wake-moment inequality $2\sum_{j=0}^{10}(-1)^j(j-5)^2o_j \geq 168$, where o_j counts odd classes at level j after translating the lowest level to zero. Equality again characterizes the standard highway. A wake-specific period bound and complete enumeration of 1,616 masks prove this extremal statement. For a prescribed wake and any full non-axis drift, a computable period bound makes inverse synthesis terminate; quadratic parity tests also give analytic exclusions. For full axis drift $(a, 0)$, represent odd classes in $0 \leq x < a$. Their strict inversion count is even, at every period and height. The axis proof eliminates unknown horizontal row circulations through a port-permutation identity and seam winding, reducing the obstruction to the order of the wake endpoints. It gives excluded families and a sparse integer-operation parity test.

Keywords. Langton's ant; finite support; periodic highway; computer-assisted proof; wake rigidity; exact enumeration.

1 Introduction and main theorem

A periodically translating ant leaves a black trail behind it. Its turn word determines that trail, but the reverse question is less clear: how much of the trail determines the word? We prove that a sparse description of the layers of the classical highway already forces its full 104-turn sequence. A second rigidity statement uses an extremal statistic instead of prescribing growth or interior sparsity: among all width-ten traces of full drift $(2, -2)$, the standard highway uniquely minimizes a signed quadratic wake moment (Theorem 8.1). A complementary question asks which trails can occur at all. Non-axis wakes have a quadratic parity character; axis wakes have an inversion law after unknown row circulations are eliminated. These conditions and the rigidity theorem use the same stabilized ordering of visits to full translation classes.

The classical ant turns right on white and left on black, toggles the current cell, and moves one lattice edge. We use Cartesian headings $\mathrm{N} = (0, 1)$, $\mathrm{E} = (1, 0)$, $\mathrm{S} = (0, -1)$ and $\mathrm{W} = (-1, 0)$, numbered $0, 1, 2, 3$ in that order. A complete finite seed lists every initially black cell; all other cells are initially white. A periodic trace repeats a turn word while restoring the heading and translating the head by a nonzero vector. Even when the initial black seed is finite, this trace leaves an unbounded wake. Cells related by integer multiples of the period's displacement belong to one *translation class*. Classes receiving an odd number of visits per period describe the black strands of the wake; classes receiving an even number contribute no persistent strand.

For displacement $(2, -2)$, each diagonal level $x + y = t$ has two translation classes, distinguished by whether x is even or odd. We count how many of those two classes are odd, rather than specifying which ones they are. A previous classification prescribed every odd class of the standard wake [5]. Here only the displacement, black growth, transverse width and a sparsity condition on the interior levels are given. The complete wake and period are consequences of these data.

The next definition makes the sparsity condition precise. Transverse width measures the range of $x + y$ visited by the trace during a period.

Definition 1.1. Fix an initial position and heading. Executing the geometric turns of a heading-restoring word $w = w_0 \cdots w_{P-1}$ gives pre-update positions $q_i \in \mathbb{Z}^2$ and $d = q_P - q_0$ its drift. For $d = (2, -2)$ the full classes modulo $\mathbb{Z}d$ have labels

$$(t, r) = (x + y, x \bmod 2).$$

Let o_t be the number of classes at level t visited an odd number of times in the period. Let $g = \#\mathrm{R} - \#\mathrm{L}$ and

$$W = \max_i(q_{i,x} + q_{i,y}) - \min_i(q_{i,x} + q_{i,y}).$$

A right turn adds one black cell and a left turn removes one, so g is the net black-cell increase over the period. The wake is *interior-thin* when $o_t \le 1$ at each level strictly between the two extrema.

Thus an interior level may be visited many times; thinness requires only that at most one of its two residue classes have an odd total visit count. For example, the standard profile has counts $(2, 1, 1, 1, 1, 1, 1, 0, 1, 1, 2)$ from its lowest to its highest level. Our hypotheses allow the location of the empty level and the occupied interior residues to vary.

Theorem 1.2 (Thin-wake rigidity)**.** *Let a word w repeat forever under the classical Langton rule from a complete finite black seed. Suppose it restores the heading, has full drift $(2, -2)$, has net black growth $g = 12$, and has transverse width $W = 10$. If its wake is interior-thin, then $P = 104$ and w is a cyclic phase of the standard highway.*

There is no period bound in this hypothesis. The seed need not occupy a prescribed box, and genuinely unread stationary debris is allowed. The theorem also applies to an eventual periodic suffix after an arbitrary finite transient. Its boundary state is again a complete finite-support state.

Corollary 1.3. *Any nonstandard finite-support periodic highway with drift $(2, -2)$, growth twelve, and transverse width ten has both longitudinal residue classes odd on at least one strictly interior transverse level.*

The full displacement is measured over the chosen word. For example, repeating the standard word twice gives displacement $(4, -4)$ and growth 24. Fixing $(2, -2)$ therefore fixes the scale of the period in the classification.

Jillhewar's transverse exclusions already give width at least ten for every diagonal finite-support periodic highway [2, Section 9, theorems "No highway of transverse width two, four, six, eight"]. Theorem 1.2 addresses uniqueness within that smallest surviving width, with the additional full-drift, growth and interior-thinness hypotheses. The general inverse problem in Appendix A starts instead from a prescribed wake and non-axis drift and computes a terminating search range.

The proof combines analytical reductions with finite integer computations. The supplied-word realization criterion is prior work [2, Section 3, "Periodic-realisability criterion"]; short-factor constraints belong to the symbolic study of the ant [1]. The specific upper certificate and ordered-event engine are retained from the standard-wake project [5]. The contribution here is the removal

of the complete-wake and missing-level hypotheses, together with the complete classification of the resulting finite family. No lower-width exclusion is used as a premise. Appendix B develops a separate necessary parity test for arbitrary-width mixed wakes, extending the attributed port-permutation factorization.

2 Finite support and stabilized visit order

A complete state is $X = (B, p, h)$, where $B \subset \mathbb{Z}^2$ is finite, $p \in \mathbb{Z}^2$ is the head position, and $h \in \{N, E, S, W\}$. Write F for one literal update and $\chi(x, y) = (-1)^{x+y}$. The quantity

$$\sigma(X) = \begin{cases} \chi(p), & h = N, S, \\ -\chi(p), & h = E, W \end{cases}$$

is invariant: each update changes both position parity and heading axis. All times below are pre-update times.

Fix a nonzero drift and write the positions in one class as $q_i = b + a_i d$. A visit to the physical cell $b + nd$ at phase i occurs, when present, at time

$$(n - a_i)P + i.$$

Thus the stabilized order of phases in that class is decreasing a_i, followed by increasing i. Equivalently, its keys are $i - Pa_i$.

Lemma 2.1 (Supplied-word criterion)**.** *A heading-restoring nonzero-drift word admits a finite-support realization if and only if every class, in stabilized order, alternates* R, L, R, L, ... *starting with* R.

Proof. Far enough along a class every phase is present, and infinitely many such physical cells are initially white. Their first turn is R; successive reads toggle the colour. This proves necessity. For sufficiency, the phases available at $b+nd$ form the suffix obtained by removing all phases with $a_i > n$. For $\min a_i \le n < \max a_i$, initially blacken this cell precisely when the number of removed phases is odd; leave every other cell white. Each available suffix starts with the correct colour and alternates. Induction in real time verifies every prescribed update. Only finitely many boundary cells have been blackened. This is the construction of [2, Section 3, "Periodic-realisability criterion"]. □

Each class therefore contributes zero or one to #R − #L, according as its cardinality is even or odd. In particular,

$$g = \sum_t o_t. \tag{1}$$

Lemma 2.2 (Extreme classes)**.** *Every visited class on a transverse extreme has exactly one event, a right turn. In the top-rail normalization the top is $t = 0$, the head faces east, $\sigma = -1$, and the bottom is $t = -10$.*

Proof. The E and N steps increase $t = x + y$; W and S decrease it. Since $\chi = (-1)^t$ is fixed on a level, the invariant σ fixes the arrival axis there. At the top, the incoming heading is N or E and the outgoing heading is S or W. Hence every visit to a fixed extreme class has the same arrival and departure directions, and thus the same turn. Lemma 2.1 forces this constant stabilized word to be the singleton R. At the top, N followed by a right turn would move E above the extreme; the admissible right turn is E to S. At the bottom it is W to N. Translate a top event to the origin. The stated gauge and width follow. □

There are at most two extreme classes on either boundary, because the quotient is by the full vector $(2, -2)$, not its primitive direction. The geometric interval is a consequence of the candidate's stated width and normalization; it is not a wall added to the original dynamics.

3 Five possible profiles, derived analytically

In the top-rail gauge ($\sigma = -1$) of Lemma 2.2, let m_t count the phase events on level t. Let u_t count straight passages from $t-1$ through t to $t+1$. Flow balance in the closed transverse walk gives the same number of reverse straight passages. At even t, right turns reverse the transverse direction and left turns continue it; at odd t the roles are reversed. Consequently

$$m_t = (-1)^t o_t + 4u_t. \tag{2}$$

The movement axis alternates, so there are $P/2$ horizontal moves. Their net displacement is two, implying that $P/2$ is even. Summing (2) gives

$$P \equiv 0 \pmod 4, \qquad \sum_t (-1)^t o_t \equiv 0 \pmod 4. \tag{3}$$

Proposition 3.1 (Profile reduction)**.** *Under Theorem 1.2's hypotheses both extreme levels have two odd classes. Exactly one interior level is empty, and it is one of*

$$h \in \{-9, -7, -5, -3, -1\}.$$

All other interior levels have one odd class.

Proof. The two rails supply at most four odd classes; the nine interior levels supply at most nine. Growth twelve forces the rail total to be three or four. If it is three, all nine interior levels are occupied. Their signed sum is $4 - 5 = -1$, so the total in (3) is two, a contradiction. Thus the rail total is four and exactly eight interior levels are occupied. If h is the empty level, the signed sum is $4 + (4 - 5) - (-1)^h = 3 - (-1)^h$. It is divisible by four exactly for odd h. □

For a fixed h, list the eight occupied interior levels in increasing order. An eight-bit mask assigns their longitudinal residues. There are 256 masks. Changing the origin to the other singleton top event swaps every residue, i.e. replaces the mask by its bitwise complement. We choose the origin so that the highest occupied interior strand has residue one, giving masks $128, \ldots, 255$. This normalization uses an actual top event and a lattice translation; it uses neither reflection nor colour complementation.

Corollary 3.2 (Standard layer profile)**.** *Suppose the full drift is* $(2, -2)$ *and, after normalizing the highest odd level to zero, the odd-class counts at levels* $-10, \ldots, 0$ *are*

$$(2, 1, 1, 1, 1, 1, 1, 0, 1, 1, 2),$$

with no odd classes elsewhere. Every finite-support periodic realization has period 104 *and the standard read word. The longitudinal residues on the eight singleton levels need not be specified.*

Proof. The supplied-word criterion gives growth twelve. Every visited extreme is odd, by Lemma 2.2, so the actual width is ten. The supplied profile is interior-thin. Apply Theorem 1.2. □

In particular, specifying all twelve standard class labels gives the complete-wake uniqueness theorem [5]. There are two proof routes from the five profiles. The direct route in Sections 4–7 checks all 76 residue assignments after count reduction. The mixed-wake parity identity of Appendix B first eliminates the profiles with empty level $-9, -5, -1$, leaving 40 assignments; Section 7.1 explains that shorter route. We present the full enumeration because its proof uses only local count and chronological conditions, independently of the port-permutation identity.

4 An all-period upper certificate

The period is bounded before any chronology search. In the top-rail normalization $\sigma = -1$, the analytic clock bound of Theorem A.2 gives

$$P \le \sum_{t=-10}^{0} (-1)^t o_t (3^{|t+5|} - 1) \le 928$$

for each of the five profiles. This already makes the inverse problem finite. The following integer graph certificate reduces the caps to 160–192, which makes exhaustive enumeration practical.

Let $\mathcal{B}$ be the supplied list of 588 impossible words, of length at most 21. Each has consecutive reads of a physical cell requesting the same turn, so it is impossible on every initial colouring. Construct contexts from the empty word and all prefixes of words in $\mathcal{B}$, retaining those with no suffix in $\mathcal{B}$. On appending a letter, reject a forbidden suffix; otherwise retain the longest suffix that is a context. This defines a deterministic suffix automaton with 4,931 contexts.

A graph vertex is (t, h, a) with $-10 \le t \le 0$ and $h \equiv t + 1 \pmod 2$, where a is a context. Place the geometric walk of a so that its endpoint has transverse coordinate t and heading h; retain the vertex exactly when the whole placed context stays in $[-10, 0]$. An edge performs one literal turn and move and applies the suffix transition, provided its target is retained. This gives 54,718 vertices and 90,158 edges.

For a periodic physical trace, each retained context is a suffix of its true history and therefore lies inside the same strip. After the finite lookback has been supplied, the context depends only on phase, so the trace lifts to a closed graph walk. Repeated vertices are allowed. This coverage statement, rather than a converse realization claim for all graph paths, is what the upper bound uses.

Set

$$\alpha_t = -2(-1)^t (t + 5)^2, \tag{4}$$

whose values from -10 through zero are

$$(-50, 32, -18, 8, -2, 0, -2, 8, -18, 32, -50).$$

The integer recipe starts all vertex potentials at zero and repeatedly raises y_w to $\max(y_w, y_v + 1 + \alpha_t \varepsilon)$ along edges. A guard at 180 makes this reconstruction a finite procedure; every edge is checked afresh at termination. The retained certificate passes all inequalities

$$y_v - y_w + \alpha_t \varepsilon \le -1, \qquad \varepsilon = 1 \text{ (R)}, \quad -1 \text{ (L)}. \tag{5}$$

Its maximum potential is 180; the verifier checks each integer edge inequality directly.

Lemma 4.1 (Profile budgets)**.** *The five profiles of Proposition 3.1 have respective all-period bounds* $192, 168, 160, 168, 192$.

Proof. Sum (5) along the actual lifted closed walk. Vertex potentials telescope. Lemma 2.1 gives the signed turn excess o_t at level t, so

$$P \le -\sum_t \alpha_t o_t.$$

Substitution of the five profiles yields the displayed bounds. □

The same graph and potentials certify all profiles. The graph forgets the longitudinal residue only for this necessary upper bound. The exact search in the next section retains both residue classes.

5 Finite residue and count reductions

Sweep the levels from top to bottom. At a class adjacent to the processed part, let I, J be the numbers of arrivals and departures on that side. Number the class events by their stable ranks. Let a count processed-side arrivals on even ranks and $o \in \{0, 1\}$ be the class's odd indicator. Let a_k and b_k indicate arrival from and departure to the processed upper side at stable rank k. The alternating turns give

$$b_k = a_k \oplus (k \bmod 2) \oplus (t \bmod 2).$$

For even t, departure-ones count as $a+(\lfloor m/2 \rfloor-(I-a))$; for odd t, they count as $(\lceil m/2 \rceil-a)+(I-a)$. Solving for m gives the two cases below. A class length m is count-feasible exactly when

$$m = \begin{cases} 2(I+J-2a)+o, & t \text{ even}, \\ 2(J-I+2a)-o, & t \text{ odd}, \end{cases} \quad m \geq 0, \quad 0 \leq a \leq \lceil m/2 \rceil, \quad 0 \leq I-a \leq \lfloor m/2 \rfloor. \tag{6}$$

Indeed the two rank parities provide precisely these available arrival slots; the local turn rule fixes departures. This is exact local count feasibility, not sufficient global chronological realizability.

For a frontier (b_0, b_1, c_0, c_1) of outgoing and incoming edge counts, the new row sees the following counts. At even levels arrivals are horizontal and switch $r = x \bmod 2$, while departures are vertical and preserve it; at odd levels the roles reverse:

$$(I, J) = \begin{cases} ((b_0, b_1), (c_1, c_0)), & t \text{ odd}, \\ ((b_1, b_0), (c_0, c_1)), & t \text{ even}. \end{cases}$$

If its lengths are m_0, m_1, the next frontier is

$$(m_0 - J_0, m_1 - J_1, m_0 - I_0, m_1 - I_1). \tag{7}$$

Start with $(1, 1, 1, 1)$ at a saturated rail and require the same frontier at the other rail.

Modulo four, the a term disappears from (6); hence the residue frontier is uniquely determined by the wake bits. It can be propagated in either direction. A wrong terminal residue rejects a mask at every period. A normalized physical event also satisfies

$$i \equiv 2r + (t \bmod 2) \pmod 4. \tag{8}$$

This follows directly by alternating vertical and horizontal unit moves. The four phase groups have $P/4$ events. Their count residues must agree, and their common residue fixes P modulo sixteen.

Proposition 5.1 (Finite input inventory)**.** *For $h = -9, -7, -5, -3, -1$, respectively, the modular tests leave $12, 20, 12, 20, 12$ normalized masks. Thus 76 wake assignments remain.*

Proof. Evaluate (6)–(7) modulo four for the 128 masks per profile, and check (8). The supplement regenerates every integer calculation in both directions and checks complementary masks. It also tests all 2304 masks before normalization and the profile parity rejection for the four even empty levels. No numerical optimizer is involved. □

For chronology pruning we also use nonnegative completion tables from (6)–(7). At each frontier retain the least cost of a counted completion. Because future count transitions depend only on the frontier, this is an exact minimum for the relaxation and a safe lower bound for a real completion. The complete table is regenerated from the wake by this finite recurrence.

6 Chronology-exact enumeration and its coverage

In each full class (t, r), list all events in stable order. Their turns are fixed: even ranks right, odd ranks left. Enumerate their incoming sides; literal turning fixes their outgoing sides. In a fixed checker sector an undirected lattice edge has one direction of use. Therefore its ordered source and target occurrences match first-to-first, with no free matching permutation. For class representatives $b_{t,r} = (r, t - r)$ the matching edge has integer voltage δ determined by

$$b_{\text{source}} + \text{outgoing unit vector} = b_{\text{target}} + \delta(2, -2).$$

Lemma 6.1 (Exact event-graph test)**.** *A fully joined event graph represents a finite-support periodic trace exactly when it forms one directed cycle with total voltage one and, along that cycle, the lifted times $j - Pv$ increase strictly with stable rank in every class.*

Proof. A real trace yields precisely such ordered edge matches and one phase cycle. Conversely the voltage identities reconstruct its lattice walk, headings and drift. The stated time order gives the alternating stabilized word of Lemma 2.1, which constructs a finite seed for the infinite repetition. A separate full-field parity calculation verifies every accepted seed. □

The following conditions prune a partial graph while preserving every possible completion. Count tables only lower-bound unbuilt cost. A premature directed cycle cannot later join mandatory unbuilt events. The four phase groups cannot exceed the cap divided by four. Within a fixed open chain, consecutive reads of the same lifted cell must have consecutive stable ranks: a missing rank cannot be inserted into a trajectory segment whose every update is already fixed.

For completeness, the period-dependent time pruning does not assume the final period equals the cap. A partial graph is a union of open directed paths. For an event u, let $c(u)$ be its path, p_u its step offset from the first event, and v_u the accumulated edge voltage along that offset. Let z_c be the unknown phase of the first event on path c. The stabilized time is

$$\tau_u = z_{c(u)} + p_u - Pv_u.$$

Consecutive class ranks require a gap at least four by (8). A simple cycle of these difference inequalities on q chain variables has weight $C + KP$ with integer K. Indeed a required rank step $u \to u'$ gives

$$z_{c(u')} \geq z_{c(u)} + 4 + p_u - p_{u'} + P(v_{u'} - v_u).$$

A simple constraint cycle uses each of its q paths once. On a path with n_c events, the two selected offsets differ by at least $-(n_c - 1)$. Thus $C \geq 4q - \sum_c (n_c - 1) \geq 5q - n$, where n events have been built. If $K \geq 1$ it is positive at every $P \geq n$; if $K = 0$ its sign is constant; if $K < 0$ it decreases with P. Thus a positive constraint cycle at the search cap excludes every admissible smaller period. All operations use exact integers.

No pruning assumes bounded physical memory or a preselected seed box. Only completed traversals are exclusions. A time or node limit is explicitly inconclusive.

6.1 A disjoint partition of the final difficult case

One case, $h = -1$ and mask 187, requires a larger traversal. Its phase residues are all two, so $P \equiv 8 \pmod{16}$; the 192 upper bound therefore reduces to a search cap of 184.

The top-down search deterministically enumerates the 673 partial graphs at the completed-layer frontier preceding row -5. Shard j of eight follows exactly the ordinals congruent to j modulo

eight. Each complete candidate crosses that frontier once. Every shard uses the same recursion and rollback rules to enumerate this cut list, so their disjoint union covers the entire search. Every shard must finish. Matching cut counts and digests are additional implementation controls, not a replacement for this partition argument.

7 Completed classification and proof of the main theorem

Table 1 gives the completed search totals. These count constructed prefixes, not distinct words or simulated ant updates. The last row includes prefixes repeated before the partition frontier in the eight shards.

Empty level	Period bound	Masks	Visited prefixes	Accepted traces
-9	192	12	56 388 171	none
-7	168	20	10 030 681	none
-5	160	12	2 353 673	none
-3	168	20	23 059 897	standard only
-1	192	12	299 233 450	none

Table 1: Completed exact enumeration. All caps are justified before the search; there is no period cutoff in the main theorem. The final difficult case uses cap 184 after its independently checked phase congruence.

All 76 canonical masks are covered by 83 completed native runs, counting the eight shards separately. The total is 391 065 872 prefix visits. The sole survivor is mask 163 with empty level -3. Its complementary mask 92 is the same geometric trace with the other top event chosen as origin.

The complete odd-class set is consequently forced to be

$$O_{\mathrm{std}} = \{(-10,0),(-10,1),(-9,1),(-8,1),(-7,0),(-6,0),\\ (-5,0),(-4,1),(-2,0),(-1,1),(0,0),(0,1)\}.$$

The normalized word, head east at the origin, is the concatenation of these two lines:

```
RRRLRLLRRLLLLRRLLRRRRLLRRLRLLRLLRLRLLRRRRLLRLLRRRRLL
RRRRLLRLRRRRLRLLLLRRRRLRRLRRRRLLLLRLRRRRLRRRRLLLLRLR
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It is a cyclic phase of the fixed standard word. A complete seed is

$$B = \{(-7,0),(-7,1),(-6,-3),(-6,-2),(-6,-1),(-6,2),\\ (-5,-2),(-5,0),(-5,1),(-5,2),(-4,-2),(-4,1),\\ (-3,-3),(-3,1),(-2,-3),(-2,-1),(-2,1),(-1,-2),(-1,0)\}.$$

Nineteen is not claimed to be the minimum seed size over other phases.

Proof of Theorem 1.2. Proposition 3.1 gives the five profiles and saturated rails. Complementary-origin normalization and Proposition 5.1 leave the 76 masks. Lemma 4.1 bounds every possible period, and the chronology-exact enumeration covers every candidate below these bounds by Lemma 6.1 and the sound pruning arguments. The completed inventory contains exactly the displayed standard word. Its separate infinite-future verification supplies existence. Therefore $P = 104$ and the wake is standard. □

7.1 A shorter parity route to the same classification

The mixed-wake identity in Appendix B removes three of the five profiles analytically. In the top-rail frame its Q layers are the odd levels: translating by $(0, 1)$ changes the checker convention and leaves the lane labels and order unchanged. If h is the missing layer, then $V = 4$ and $J = (h + 37)/2$. The condition $J + V/2 \equiv 1 \pmod 2$ leaves precisely $h = -7, -3$. It is independent of all eight interior residue choices. For $h = -7$, the twenty retained normalized chronological exclusions at cap 168 apply. The $h = -3$ case is the standard layer-profile theorem, including its complete residue classification. This yields an alternative proof using those two retained subcertificates. The full five-profile computation above remains a complete route with the same conclusion.

More generally, put the full extreme rails at $-2k, 0$ and one odd class at each interior level except $h = -2k + 1 + 2j$. Heading restoration forces k odd. Then $V = k - 1$ and $J = (k + 2)(k - 1)/2 + j$, so the identity requires j odd, equivalently $h \equiv 1 \pmod 4$. This derivation supplies the necessary single-gap parity test at every such width.

8 A sharp width-ten wake moment

Normalize the lowest transverse level to zero, and write o_j for the number of odd full-drift classes at level j, $0 \le j \le 10$. This is a translation of the earlier top-zero convention; longitudinal lanes remain distinct.

Theorem 8.1 (Sharp width-ten moment)**.** *Every complete finite-support periodic realization with full drift* $(2, -2)$ *and transverse width* $W = 10$ *satisfies*

$$\mathcal{K} = 2\sum_{j=0}^{10}(-1)^j(j-5)^2 o_j \ge 168.$$

Equality holds exactly for the standard period-104 turn word, up to cyclic phase and the declared lattice normalization.

The theorem allows every growth, interior multiplicity and finite initial field compatible with these geometric hypotheses. The proof derives its own period bound separately for each wake. The standard profile attains 168.

8.1 Finite-support colour chronology

A phase i in the drift class of q_i can be written $q_i = b + a_i d$. At the physical cell $b + nd$, for sufficiently large n, all phases of that class are present. Their time order is by

$$\tau_i = i - Pa_i. \tag{9}$$

Because the input black support is finite, such a far-ahead cell starts white. Consecutive visits flip it. Hence the letters in every drift class, ordered by (9), must alternate $R, L, R, L, \ldots$ starting R. Conversely, the boundary colours at the finitely many truncated classes can be chosen to realize any geometric periodic word obeying these alternations; this exact criterion is due to Jillhewar [2] and is also derived in Lemma 2.1.

In particular, the signed excess $R - L$ in one drift class is 0 or 1; it equals the indicator of an odd class. Thus

$$g = \#R - \#L = \sum_j o_j \equiv 0 \pmod 4. \tag{10}$$

The end rails of a translating trace are geometric turning points. There all visits to a given class must have the same required turn; alternation forces exactly one white-first right turn. Consequently the extremal occupied levels are also extremal odd-class levels and the incoming axis is horizontal. Translation and the top-class phase fix a normalization with levels 0 and 10, and two longitudinal lanes $r = 0, 1$.

Let c_r count odd classes on even levels of lane r, and q_r those on odd levels. The signed mod-four residue restrictions and the two-lane mixed-wake interleaving identity of Appendix B give

$$c_0 - q_0 \equiv c_1 - q_1 \equiv c_0 - q_1 \equiv c_1 - q_0 \equiv 2 \pmod 4, \tag{11}$$

$$V = q_0 + q_1 \equiv 0 \pmod 2, \qquad J + V/2 \equiv 1 \pmod 2, \tag{12}$$

where J counts ordered pairs of an odd class on an odd level and an odd class on a higher even level. These are necessary tests, never a sufficient realizability test. Previously established finite-support restrictions proved in the attributed residue analysis of Appendix B ensure $g \in \{8, 12, 16\}$ at width ten; these restrictions reduce the possible growth without prescribing it as input. For completeness, positive growth is divisible by four and there are at most 22 full classes. Growth four is excluded by the finite-support restrictions summarized in Appendix D. At growth twenty, the residue conditions force six occupied even levels and four occupied odd levels in each lane: all even levels then have both classes. Each odd class has an even number of even classes above it, so J is even; also $V = 8$. This contradicts (12), leaving exactly the three stated growth values.

8.2 An all-period bound from the wake

The residue-independent forbidden-factor graph for this width contains $54,718$ vertices and $90,158$ directed edges. Its integer certificate consists of vertex potentials y_v with

$$y_v - y_{v'} + \alpha_j \epsilon \le -1, \qquad \alpha_j = -2(-1)^j (j-5)^2, \tag{13}$$

for each edge $v \to v'$ from transverse level j, where $\epsilon = +1$ on an R and -1 on an L turn. Every physically realizable periodic word of width ten lifts to a closed walk in this graph. Importantly, the argument covers walks with repeated vertices; it is not a simple-cycle-only bound. Section 4 constructs this graph, proves coverage and gives the integer inequalities. Those arguments do not assume a particular odd wake.

Summing (13) over a true period eliminates the y terms. The stable-class alternation makes the total signed turn contribution at level j equal to o_j. Therefore

$$P \le -\sum_{j=0}^{10} \alpha_j o_j = \mathcal{K}. \tag{14}$$

This is an effective, wake-specific cutoff *derived from*, rather than added to, the hypotheses. If $\mathcal{K} < 4$, no nonempty heading-restoring candidate exists. To prove Theorem 8.1 it remains to exhaust all permitted wakes with $4 \le \mathcal{K} \le 168$.

8.3 Complete finite reduction

8.3.1 Wake masks and exact crossing-count relaxation

Encode the odd set of each lane by an 11-bit mask a, b. There are exactly 496 signed-residue-feasible masks for one lane. Applying both extreme-level conditions, (11), (12) and (10) yields $13,216$

Growth g	$\mathcal{K} \le 168$ reps	Realizations
8	371	0
12	1,020	1
16	225	0
Total	1,616	1

Table 2: Complete bounded-moment realization search; the middle column counts full wake masks, not trajectories. The solution column counts accepted normalized words, verified to be one standard P104 phase.

normalized pairs. We do not bound a black source's coordinates, nor choose a maximum length of a periodic word.

At one full class of transverse parity $j \bmod 2$, let I, J count its incoming and outgoing ports on the already processed side. If A of the I arrivals occupy even stable visit ranks and the class has odd indicator $o \in \{0, 1\}$, its total visit count is exactly

$$m = 2(I + J - 2A) + o \qquad \text{on even } j, \tag{15}$$

$$m = 2(J - I + 2A) - o \qquad \text{on odd } j, \tag{16}$$

with the necessary bounds

$$0 \le A \le \lceil m/2 \rceil, \quad 0 \le I - A \le \lfloor m/2 \rfloor, \quad m \ge 0. \tag{17}$$

Matching port counts across adjacent layers supplies a four-integer boundary. The dynamic program retains the minimum cumulative event cost at each complete boundary, growth and horizontal displacement. Because future count constraints use only these data, keeping the cheapest partial count history is exact for this *necessary relaxation*; no chronological realizability is inferred from a count survivor.

All $13,216$ necessary masks are processed. Exactly $2,790$ pass the complete integer port relaxation. When both top classes are present, choosing the other top visit as origin exchanges longitudinal residues, yielding exactly 876 paired duplicate masks. Thus $1,914$ canonical representatives remain. Of those, $1,616$ have $\mathcal{K} \le 168$ (Table 2).

8.3.2 Physical chronology and completeness

The count relaxation is not enough. The second stage attaches all actual visit events in each full class in stable-rank order. Each kth ranked visit has turn R if k is even, L otherwise; its entering direction determines its leaving direction. Directed-edge occurrences are matched first-to-first in the stable order, as their lifted times differ by a fixed $1 - P \cdot$ voltage along a directed quotient edge. Each partial object is therefore a union of directed chains. A premature closed component is rejected, since it cannot later join the unique complete periodic cycle.

Let z_A be a chain-root time, p_i the position of event i in chain A and v_i its accumulated displacement voltage. Its stabilized time is $z_A + p_i - Pv_i$. For consecutive stable ranks i and j on different chains A, B, physical non-overlap requires

$$z_B - z_A \ge \max\{4 + p_i - p_j, \ell_A\} + (v_j - v_i)P, \tag{18}$$

where ℓ_A is the length of chain A. For ranks in the same chain one checks the direct gap four, not the cross-chain length. Each cross-chain edge in (18) has an integer constant term at least $\ell_A > 0$.

Along a cycle of difference constraints the total bound has form $C + KP$ with $C > 0$. When $K \geq 0$ positivity precludes every positive period; when $K < 0$ positivity at an upper cutoff L implies positivity for all $P \leq L$. Thus testing a positive constraint cycle at the proved cap $L = \mathcal{K}$ cannot wrongly reject a smaller real period. The within-chain case follows by its direct monotonicity in P.

An incrementally maintained difference system applies this pruning after each port addition and supports exact rollback. A separately constructed root-variable constraint oracle rebuilds the entire partial problem and checks the new incremental algorithm on selected streams. This is a scoped implementation audit, not a replacement for the mathematical coverage argument or for the complete enumeration.

At a fully matched object, acceptance requires *one* directed phase cycle, full voltage $+1$, horizontal drift $+2$, strictly increasing stable ranks within every class, and exact heading restoration. Following the cycle recovers a geometric turn word. Conversely, every realizable periodic trace with the supplied odd mask creates exactly such a ranked event object; all enumeration and pruning rules above preserve it. The entire future of an accepted word is checked separately using complete finite-support parity rather than inferred from a repeated finite prefix.

8.3.3 Finite certificate and equality

The machine-readable certificate lists the mask, growth, individual cap, number of explored prefixes, and number of accepted words for *every* $\mathcal{K} \leq 168$ canonical representative. All $1,616$ searches terminated with status `COMPLETE`. The accepted list has exactly one period-104 word, with the standard 12 odd classes. The aggregate explored-prefix count is $767,594,212$; it is a code-coverage count, not an ant update count.

An independent ray-parity checker constructs a complete nineteen-black-cell boundary source for that word and verifies its first required colour at each phase. It tests the finitely many old-black-hit epochs, all self-write parity thresholds and the final unbounded epoch interval. All 194 required phase/epoch checks pass. This establishes the positive realization and the value $\mathcal{K} = 168$.

Every candidate with $\mathcal{K} < 168$ has no accepted word. At equality the only accepted word is the standard phase. The cover and the derived cutoff exclude any other periodic realization in the declared class at or below this moment value. This proves Theorem 8.1. □

9 Axis-wake inversion parity

We use the complete-state and pre-update conventions of Section 2.

We study a nonempty turn word $w = w_0 \cdots w_{P-1}$ whose infinite repetition is executed from such an input. Its heading returns after P updates, and its *full* displacement is a nonzero axis vector. A quarter-turn rotation normalizes it to $d = (a, 0)$ with $a > 0$. The word need not be a least period. The complete colouring need not translate: it retains all emitted wake and unread debris. A finite transient before the repeated suffix is allowed, since the field at its end is still finite.

Let q_i be the pre-update position of phase i. Translation classes are cells modulo $\mathbb{Z}(a, 0)$, represented by $(x \bmod a, y)$. Let O be the finite set of classes containing an odd number of the P phases, and let $g = |O|$. Define the *strict inversion count*

$$I(O) = \#\{(u, v) \in O^2 : x_u < x_v,\ y_u > y_v\}. \tag{19}$$

Equal rows or columns do not contribute. Equivalently, sort O by increasing (y, x) and count strict inversions of the resulting x list.

Theorem 9.1 (Axis-wake obstruction)**.** *Every finite-support periodic realization with full displacement $(a, 0)$, $a > 0$, has even a and satisfies*

$$\boxed{I(O) \equiv 0 \pmod 2.}$$

There is no bound on P, vertical span, g, even visit multiplicities, initial black cardinality or initial coordinates.

The test is intrinsic to the cylindrical wake. Translating the coordinates, changing the phase origin or changing representatives does not alter its parity on the relevant domain.

Relation to earlier results. Jillhewar's stabilized-class criterion, signed mod-four residues and special-axis lifted-rank factorization supply the underlying methodology [2]. The non-axis wake character in Appendix C uses uniqueness of finite horizontal and vertical boundary antiderivatives. That uniqueness fails on an axis cylinder: a whole horizontal row can be added without changing any endpoint. The new step here is to prove that these row cycles lie in the radical of the inversion polarization and to eliminate their remaining scalar contribution using winding. A second step reduces the resulting character to the endpoint order (19). For $a = 2$ the result recovers the existing growth divisibility condition; the new scope includes every larger even full axis displacement.

9.1 Stable visits and the cylindrical parity graph

Write positions in a fixed translation class as $q_i = b_v + k_i d$. A visit to the physical cell $b_v + nd$ at phase i occurs at time

$$nP + \tau_i, \qquad \tau_i = i - Pk_i,$$

provided $n \geq k_i$. For sufficiently large n all phases occur, and the cell was initially white. Thus increasing τ_i orders the class visits as $\mathrm{R}, \mathrm{L}, \mathrm{R}, \mathrm{L}, \ldots$. Different phases have different keys, since equality would imply equality modulo P. This is the necessary part of the periodic-realisability criterion of [2].

The signed excess $D_v = \#\mathrm{R} - \#\mathrm{L}$ in a class is zero or one, and is exactly its odd indicator. Hence the total excess is g. Heading restoration gives $4 \mid g$. It also makes P even. Since horizontal and vertical moves alternate and the net vertical displacement is zero, $P/2$ is even. Therefore $4 \mid P$, and the horizontal net displacement a is even.

The product of checker parity with the sign distinguishing horizontal and vertical incoming headings is invariant. Label vertical-arrival vertices Q, horizontal-arrival vertices C, choosing the coordinate gauge in which Q has even $x + y$. Horizontal directed edges run from Q to C and vertical edges from C to Q. Each visit uses one edge of each axis. Consequently the horizontal and vertical boundaries of the traversed-edge indicator modulo two are both $D = \mathbf{1}_O$. Every row and every column of O has even population.

Pair consecutive y endpoints on each column to obtain the unique finite vertical edge set with boundary D. On each row choose the horizontal seam edge from column $a - 1$ to column zero to be absent, and pair consecutive x endpoints in $0, \ldots, a - 1$. Call the union of these horizontal and vertical interval edges $G_0(O)$. All edges retain their physical direction and port. For $a = 2$, the east and west edges between the same quotient vertices remain distinct parallel edges.

Lemma 9.2 (All possible parity lifts)**.** *The actual parity graph of any period with wake O is*

$$G = G_0(O) \oplus \bigtriangleup_{y : h_y = 1} R_y, \tag{20}$$

where R_y is the full horizontal row cycle and only finitely many h_y are nonzero. Moreover $\sum_y h_y \equiv 1 \pmod 2$.

Proof. A finite boundary-free edge set on a vertical line is empty. On a horizontal cycle it is either empty or the full cycle. Subtracting G_0 from the actual edge indicator gives (20). There is no assumption that the added cycles are between the highest and lowest wake endpoints.

Lift the closed quotient walk to the plane. Its full displacement is $(a, 0)$, so its signed crossing number of the seam is one. Modulo two signs disappear. The seam-zero graph contributes zero and each R_y contributes one. This gives the asserted parity. Even traversal multiplicities do not affect it. □

9.2 The one-tour port sign

Order the finite set of relevant vertices by any total order, and at each vertex put the negative port before the positive port. Positive means movement N or E, for both incoming and outgoing ports. For a finite directed parity edge set G, let $E(G)$ be the inversion parity between the list of its edges ordered by source port and the same edges ordered by target port. Let $N_+(G)$ count its positive edges. Define

$$q(G; O) = E(G) + N_+(G) + |O \cap C| \pmod 2. \tag{21}$$

We suppress O when it is fixed. The next factorization extends the special-drift port argument in [2]; we include its derivation.

Lemma 9.3 (Actual-period character)**.** *The parity graph of every period under consideration satisfies $q(G) = 1$.*

Proof. At a vertex with m stable-ranked events, record their positive incoming and outgoing bits as α_k, β_k, $0 \le k < m$. Put $A = \sum \alpha_k$, $B = \sum \beta_k$ and $p = A + B \pmod 2$. For any binary list,

$$\mathrm{inv}(\alpha) = A(m-1) + \sum_k k\alpha_k + \binom{A}{2} \pmod 2. \tag{22}$$

Indeed, a one at position k sees $m - 1 - k$ later positions, from which its later ones are subtracted; the latter count sums to $\binom{A}{2}$.

Write K for the sum of the stable ranks of all left turns, modulo two, and I_{port} for the sum of incoming and outgoing binary inversions. At Q a left turn means unequal port bits, whereas at C it means equal bits. Applying (22) to both lists gives local corrections $p(m-1)$ at Q and $p(m-1) + \binom{m}{2}$ at C. The positive-edge binomial terms cancel globally, because a directed edge has equal source and target counts. At C one also has

$$D = 2(A + B) - 4\sum_k \alpha_k\beta_k - m, \qquad m \equiv 2p - D \pmod 4.$$

Using $D \in \{0, 1\}$ and $m \equiv D \pmod 2$ therefore gives

$$K = I_{\mathrm{port}} + L, \qquad L = \sum_Q p(D+1) + \sum_C D(p+1) \pmod 2. \tag{23}$$

A quotient edge of voltage ν changes stable time by $1 - P\nu$. This same constant applies to all its occurrences, including the period seam. Source and target occurrences therefore match first to first.

Sorting outgoing ports, transferring edge blocks and unsorting incoming ports factors the phase successor. That successor is one even-length P-cycle, even when the chosen word is a power. Hence

$$I_{\mathrm{port}} + E(G) = 1 \pmod 2. \tag{24}$$

Exchanging blocks of sizes r, s contributes rs modulo two, so the edge-block sign depends only on edge-count parities, precisely as in the definition of $E(G)$.

It remains to simplify L. In G_0, a paired horizontal or vertical interval contributes positive-incidence parity $1 + c(u) + c(v)$ at its endpoints, where $c = 0$ on Q and 1 on C. There are $g/2$ intervals of each axis. Each endpoint checker bit is counted twice; thus $\sum_{v \in O} p_v = 0$. Adding R_y changes p_v once at every vertex on that row. The row population of O is even, so the same cancellation holds for every graph (20). Furthermore $\sum_Q p = N_+$: each positive edge has exactly one Q endpoint. Expanding (23) gives

$$L = N_+(G) + |O \cap C| \pmod 2.$$

Every left turn in a stable alternating class has odd rank, so $K = \#\mathrm{L} = (P - g)/2 \equiv 0 \pmod 2$, since $4 \mid P, g$. Equations (23) and (24) now give $q(G) = 1$. □

9.3 Eliminating unknown row circulations

For a fixed edge universe and vertex order, E is a quadratic polynomial over $\mathbb{F}_2$ in the edge indicators: each inverted pair contributes the product of its two indicators. Its polarization

$$\mathcal{B}(G, H) = E(G \oplus H) + E(G) + E(H)$$

is bilinear. The following lemma is the axis-specific cancellation.

Lemma 9.4 (Row-cycle radical)**.** *For every finite edge set G and every full horizontal row cycle R_y,*

$$\mathcal{B}(G, R_y) = 0, \qquad E(R_y) + N_+(R_y) = 1 \pmod 2.$$

The first assertion does not require a boundary condition on G.

Proof. Every source and target degree of R_y is even. Swapping vertices in the common order therefore leaves $\mathcal{B}(G, R_y)$ unchanged. Place its row vertices first, ordered by column, and use bilinearity to reduce to a single edge e.

An edge off the row sees only complete source and target blocks of R_y, each of size two, so its mixed inversion count is even. A vertical edge touching the row has its row source at C, where R_y has no source block, or its row target at Q, where R_y has no target block. The remaining whole blocks again have size two. Finally a horizontal edge on the row is itself in R_y. Its inversion count with the other row edges has parity equal to the sum of its zero-based source and target indices. Both indices have the parity of its port bit, so their sum is even. This also covers the two parallel ports when $a = 2$. Thus the polarization vanishes.

Put $a = 2m$. With the row order just chosen, one sign of horizontal port matches identically among m slots and the other matches by a cyclic shift among m slots. Its sign is $m - 1$ modulo two. There are m positive edges. Their sum is $2m - 1 \equiv 1$. □

Since edge-number parity is linear, toggling a row cycle changes q by one. Lemma 9.2 and Lemma 9.3 consequently imply

$$q(G_0(O); O) = 0. \tag{25}$$

The entire unknown set of row circulations has disappeared. In particular, no bound on its support or on the number of even visits was needed.

9.4 Evaluating the character from endpoints

The next statement is purely combinatorial, on a larger domain than that of realizable wakes.

Theorem 9.5 (Endpoint identity)**.** *For even $a > 0$ and any finite $O \subset \{0, \dots, a-1\} \times \mathbb{Z}$ with even populations on every row and column,*

$$q(G_0(O); O) = I(O) + |O|/2 \pmod 2. \tag{26}$$

It holds for either checker/arrival-axis convention.

Proof. At every vertex the incoming and outgoing degree parities of G_0 are both D_v. Exchanging vertices v, w changes E by $D_v D_w + D_v D_w = 0$. We may therefore put every Q vertex before every C vertex, using row-major order within each checker colour.

Every horizontal/vertical edge pair is inverted: its sources are in the order Q, C and its targets in the order C, Q. The parity of the horizontal edge count is $|O \cap Q|$, and that of the vertical count is $|O \cap C|$. The two endpoint populations have the same parity because $|O|$ is even. Thus the mixed inversions cancel the checker term in (21), leaving

$$q(G_0) = (E_H + N_H) + (E_V + N_V). \tag{27}$$

Consider a paired horizontal interval $[l, r]$. Its inversions are exactly its internal C vertices, where the two incoming edges reverse port order. Its positive edges are counted by Q vertices from l inclusive to r exclusive. Summing modulo two gives $c(r)$. Distinct horizontal intervals have no mutual inversions in row-major order. Therefore

$$E_H + N_H = \sum_{\text{horizontal pairs}} c(\text{right endpoint}).$$

To evaluate the vertical part, change to column-major order within each checker colour. A paired vertical interval $[b, t]$ contributes $1 + c(t)$: its inversions are its internal Q vertices and its positive edges start at C vertices from b inclusive to t exclusive. Distinct paired vertical intervals have no mutual inversions in column-major order. Changing the order back adds J_{same}, the strict inversion count among endpoint pairs of the same checker colour. This follows by block exchanges: vertical source degrees at C and target degrees at Q have parity D_v. Together with (27), this gives

$$q(G_0) = |O|/2 + J_{\text{same}} + \sum_{\text{horizontal pairs}} c(r) + \sum_{\text{vertical pairs}} c(t). \tag{28}$$

Let $r(v)$ and $s(v)$ be the zero-based ranks of endpoints in row-major and column-major order. Because every row and column has even population, paired right endpoints are precisely the odd row-major ranks, and paired top endpoints precisely the odd column-major ranks. The last two sums in (28) equal $\sum_v c(v)(r(v) + s(v))$ modulo two. The parity of an endpoint's rank change is the number of endpoints with which it exchanges order. Each inverted pair u, v therefore contributes $c(u) + c(v)$ to this sum. The sum is exactly J_{mixed}, the count of inversions with unlike checker colours. Adding J_{same} proves (26). The same argument works after exchanging the checker labels. □

Proof of Theorem 9.1. Section 9.1 gives even a, even fiber populations and $4 \mid g$. Equation (25) and Theorem 9.5 give $0 = I(O) + g/2 = I(O) \pmod 2$. □

Translation and phase invariance. Vertical translation does not change row order. To move the horizontal seam, it suffices to move one column from the start of the column order to its end. If this column has n endpoints, the number of strict comparisons reversed is

$$n(g-n) - \sum_{v \text{ in that column}} (\#\text{endpoints in row } y_v - 1).$$

Both n and g are even; each row population is even, and there are an even number n of terms in the sum. The displayed number is even. Hence cyclic column shifts preserve I modulo two. Changing the phase origin translates the odd classes: visits shifted across the period seam differ by full copies of d and remain in their classes. Normalizing representatives also leaves the same set. These observations prove the asserted invariances without a geometric cutoff.

Quadratic structure. On the vector space of finite indicators with even row and column populations, $O \mapsto G_0(O)$ is linear: in each line or cut cycle, cumulative boundary parity is a linear antiderivative. Thus $q(G_0(O); O)$ is a quadratic form. Its polarization is bilinear. Addition here is symmetric difference of indicators, not superposition of physical ant trajectories.

9.5 A fast screen and excluded families

Given g explicit class representatives in binary, reduce their horizontal coordinates modulo a, deduplicate and sort by (y, x). Merge-sort inversion counting computes I in $O(g \log(g+1))$ comparisons and arithmetic operations and $O(g)$ space. Equal column values must be treated as noninversions. The integers have polynomial bit size in the input coordinates and $\log(g+1)$. The algorithm never expands a vertical gap.

For comparison with existing arithmetic screens we use the signed residue conditions from [2]:

$$\sum_{(x,y)\in O:\, x=r} (-1)^{x+y} \equiv 2 \pmod 4, \qquad \sum_{(x,y)\in O:\, y=s} (-1)^{x+y} \equiv 0 \pmod 4. \tag{29}$$

They imply $g \geq 2a$, and impose even populations on each fiber. For clarity, their mod-four potential can be written explicitly. For arbitrary $f : \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, choose S with $S(x+1) - S(x) = f(x) + f(x+1)$ and put

$$F_N = -\chi S(x), \quad F_E = \chi(S(x) - f(x)), \quad F_S = -\chi(S(x) + 2f(x)), \quad F_W = \chi(S(x) + f(x)),$$

where $\chi = (-1)^{x+y}$. Direct substitution in the eight turn/heading cases shows that $\sum_{z\in B} \chi(z) f(z_x) + F_h(p)$ is invariant modulo four. For a-periodic f, its head monodromy along $(a, 0)$ is $2\sum_{r=0}^{a-1} f(r)$. The field increment is $\sum_{v\in O} \chi(v) f(v_x)$ by stable alternation. Choosing residue indicators gives the first condition in (29). Rotating the formula and using zero vertical displacement gives the second, for arbitrary single-row indicators. This also specifies their exact-row scope.

Theorem 9.6 (Staircase exclusions)**.** *For every even $a \geq 2$ and every list of strictly increasing heights $y_0 < \cdots < y_{a-1}$ of a single parity, define*

$$O_a = \bigcup_{j=0}^{a-2} \{(j, y_j), (j+1, y_j)\} \;\cup\; \{(0, y_{a-1}), (a-1, y_{a-1})\}.$$

These $2a$ classes satisfy (29)*, $4 \mid g$ and $g = 2a$, but no finite-support periodic trace of full drift $(a, 0)$ has this wake.*

Proof. Every column has two endpoints of the same checker sign, giving $\pm 2 \equiv 2$ modulo four. Each row has two endpoints of opposite sign, giving zero. The row-major column list is

$$0, 1,\ 1, 2, \ldots,\ a-2, a-1,\ 0, a-1.$$

Only the penultimate zero causes strict inversions. Exactly $2a-3$ preceding entries are positive. This is odd, contradicting Theorem 9.1. □

For $a = 4$, heights $0, 10, 20, 30$ give eight endpoints and $I = 5$. For contrast, all four columns on rows zero and twenty give $I = 6$ and pass both the old conditions and the new screen. Passing establishes only `NOT_EXCLUDED`, not a realizable periodic word.

At $a = 2$, even row population means that every occupied row contains both classes. If there are m occupied rows, $g = 2m$ and $I = \binom{m}{2}$. Since $4 \mid g$, m is even, so $I \equiv g/4 \pmod 2$. The theorem recovers $8 \mid g$, an already established two-column consequence of the attributed rank argument. It does not claim a new proof of the separate condition $P \equiv 4 \pmod 8$.

9.6 An exact rejection count in a factorial family

Fix even $a \geq 4$ and a strictly increasing same-parity heights. Let the cycle edges on the columns be

$$E_j = \{j, j+1\}\ (0 \leq j < a-1), \qquad E_{a-1} = \{a-1, 0\}.$$

Assign these a distinct edges in any order to the a rows, placing endpoints at the assigned columns. There are $a!$ distinct labelled wakes. Each column still has degree two, and each row contains opposite checker signs, so all these wakes satisfy the conditions of Theorem 9.6 before its inversion test.

Let π_j be the row rank assigned to edge E_j. Define

$$D_{\mathrm{cyc}}(\pi) = \sum_{j=0}^{a-1} \mathbf{1}_{\{\pi_j > \pi_{j+1 \bmod a}\}}.$$

Let $A(n, k)$ be the Eulerian number counting permutations of n elements with k ordinary descents.

Theorem 9.7 (Cycle-edge parity and counting)**.** *For this family,*

$$I(O_\pi) \equiv D_{\mathrm{cyc}}(\pi) \pmod 2.$$

The number excluded by Theorem 9.1 is exactly

$$N_a = a \sum_{k \text{ even}} A(a-1, k). \tag{30}$$

This counts labelled wake assignments rejected by the screen, not all unrealizable wakes and not classes modulo symmetry.

Proof. Exchange two adjacent rows, carrying column pairs E_i, E_j. Exactly $4 - |E_i \cap E_j|$ unequal-column comparisons reverse. The inversion parity therefore changes by $|E_i \cap E_j|$. For $a \geq 4$ this is one precisely when the two cycle edges are adjacent. Starting from the staircase order, whose inversion count is $2a-3$, and sorting any row permutation by adjacent exchanges gives

$$I(O_\pi) \equiv (2a-3) + \sum_{j=0}^{a-2} \mathbf{1}_{\{\pi_j > \pi_{j+1}\}} + \mathbf{1}_{\{\pi_0 > \pi_{a-1}\}}.$$

The final indicator is complementary to the closing cyclic descent. The odd constant cancels, proving the parity formula.

Rotate the list $(\pi_0, \dots, \pi_{a-1})$ until its smallest entry is first. Every rotation orbit has a members because its entries are distinct. The first comparison is an ascent and the closing comparison a descent. The remaining $a-1$ entries form an arbitrary permutation with, say, k ordinary descents, so $D_{\mathrm{cyc}} = k+1$. There are $aA(a-1,k)$ such lists. Odd cyclic descent count corresponds exactly to even k, giving (30). The recurrence

$$A(n,k) = (k+1)A(n-1,k) + (n-k)A(n-1,k-1), \qquad A(1,0) = 1,$$

follows by inserting the largest entry into a permutation: there are $k+1$ slots preserving k descents, and $n-k$ slots increasing $k-1$ to k. Thus the displayed count is effective for every a. □

Full displacement	Labelled wakes	Excluded	Not excluded
$(4,0)$	24	8	16
$(6,0)$	720	408	312
$(8,0)$	40320	19072	21248

The heights may have arbitrary gaps. Hence each rejected assignment supplies an unbounded family of all-period exclusions.

10 Evidence and reproducibility

The core theorem is computer-assisted: its logical coverage is a written mathematical argument and its finite claims are checked by integer programs. The supplemental source hashes identify the executed programs and the complete run records identify the finite exclusions.

The accompanying source package contains the computational supplement. Its public project is `https://github.com/kehao95/langtons-ant-core/tree/main/wake_synthesis`. Within `wake_synthesis/`, `sources.json` lists the distributed files and their SHA256 hashes; `evidence/profile_classification/manifest.json` identifies the classification programs and run evidence, and the adjacent `summary.json` and `searches.json.gz` record all 76 cases and 83 jobs. The entrypoint is `wake_synthesis/reproduce.py`, run from the repository root. The manifest identifies the thin-wake computation separately from the standard-wake archive [5] and the inverse-synthesis archive [4].

The supplement contains the count routines, both ordered-event traversal directions, the local rank check, the upper-bound construction and complete computational certificates. A regenerated count implementation is cross-checked with the retained one. The small shard controls compare their union with an unsplit positive search. The full replay checks every case, all eight final shards, and the unique finite seed. Single active-cell deletions and a far-ahead blocker are rejected by the independent infinite verifier.

```
python3 wake_synthesis/reproduce.py --jobs 4
python3 wake_synthesis/reproduce.py --quick
```

The second command explicitly skips the 76-case exclusion replay, while checking its upper certificate, count tables and controls. The default is full replay, including the periodic-growth and mixed-wake checks. The option `-extensions-only` replays the arithmetic modules, including the general quadratic character, analytic unit-diagonal regressions and the thin-profile parity reduction, while retaining the classification evidence. Python's standard library and a C++17 compiler are sufficient;

no linear optimizer, SAT solver or network access is required. Every exclusion in the classification is supported by a completed traversal.

10.1 Exact all-future seed verification

For a proposed word of period P, let q_i be its phase-i position, d its nonzero drift and B its complete finite initial black set. At epoch $n \geq 0$, the colour read at phase i is

$$c_i(n) = \mathbf{1}_B(q_i + nd) \oplus \bigoplus_{\substack{0 \leq j < P,\ s \in \mathbb{Z} \\ q_j - q_i = sd \\ s>0 \text{ or } (s=0,\ j<i)}} \mathbf{1}_{\{n \geq s\}}.$$

Indeed the earlier visit at phase j has epoch $n - s$; the displayed condition is exactly its precedence over epoch n, phase i. The required colour is $\mathbf{1}_{\{w_i = \mathrm{L}\}}$. For each i, only finitely many thresholds s and initial-black epochs occur. Probe epoch zero, every threshold, every initial-black epoch, and the epoch immediately after each initial-black epoch. The formula is constant between these probes and beyond the last one, so this finite test proves every future read. The independent implementation is `verification/profile_classification/verify_word.py`, function `forever`, in the pinned supplement.

The axis module checks exact port and endpoint identities on 33,312 closed patterns, 3,491 heading-restoring geometric words and independently constructed parity controls. A separate enumeration of 41,064 cycle-edge assignments agrees with the proved Eulerian recurrence. Run `python3 wake_synthesis/reproduce.py -axis-only` for this module. These finite controls test the implementation; the radical, winding and endpoint arguments establish the unrestricted parameter claim. The checker sources and deterministic reports are bound by `sources.json`.

10.2 Reproducing the extremal-moment classification

The moment module reconstructs every necessary wake mask and the exact integer count relaxation. Its full mode runs all 1,616 chronology searches at their individual proved caps, rejects any incomplete search, and independently validates the entire infinite future of the accepted word. The source package pins the graph certificate shared with Section 4 and the source and frozen per-case outputs of this enumeration. From the repository root:

```
python3 wake_synthesis/reproduce.py --moment-only --full
```

The default moment mode checks the masks, four representative searches and the positive infinite tail. The distinction between that quick check and a complete replay is explicit in the command and its output.

11 Limitations

The moment theorem fixes full drift and width. Wakes with moment strictly greater than 168 are outside its extremal classification. Axis parity is a necessary condition; it does not construct a realization of a passing wake. Terminating inverse synthesis retains its non-axis hypothesis. The sparse axis screen counts integer operations, with coordinate bit costs additional. No primitive-direction quotient may replace the full displacement.

The theorem reduces the remaining periodic problem: a nonstandard highway retaining drift $(2, -2)$, growth twelve and width ten must have a thicker interior wake. It remains necessary

to determine whether such wakes exist, whether different numerical invariants are possible, and whether all finite seeds eventually become periodic. Those questions are not answered by the present classification. The inverse-synthesis bound and quadratic wake reconstruction apply to non-axis drift. In the moment construction an axis-normal coordinate vanishes, preventing a positive κ; in the quotient graph, axis drift permits boundary-free cycles. These methods therefore leave axis synthesis open. The parity screens are necessary conditions: a passing wake still requires chronological realization and the supplied-word criterion. The finite period cutoff ensures termination of synthesis, rather than a polynomial search-time bound.

A Wake-dependent period bounds and inverse synthesis

The standard-wake classification is one instance of a general inverse problem. Prescribe a non-axis drift d, an initial heading, and a finite set $O \subset \mathbb{Z}^2/\mathbb{Z}d$ of odd classes. We seek a periodic word and finite seed with exactly these data. The wake-dependent bounds [4] determine a finite search range directly from the input wake.

Lemma 2.1 gives $\#\mathrm{R} - \#\mathrm{L} = |O| =: g$. Heading restoration implies $g \equiv 0 \pmod 4$, P even and $d_x + d_y$ even. For $d = (m, -m)$, the complete class label is $(x + y, x \bmod m)$. The checker-sector invariant σ is determined by the supplied pose.

A.1 An integer clock for diagonal budgets

After rotating a diagonal drift parallel to $(1, -1)$, the transverse coordinate $x+y$ and heading return after each period. A clock using these quantities has a cancelling head correction: its period gain comes from the wake. We recall the unit increment identity of [3] before computing that gain.

Fix an integer c, set $s = x + y - c$, and put

$$f(s) = 3^{|s|} - 1, \qquad \epsilon(h) = \begin{cases} +1, & h = N, E, \\ -1, & h = S, W, \end{cases} \qquad G(s, e) = \begin{cases} \frac{1}{2} f(s), & es \geq 0, \\ -\frac{3}{2} f(s), & es < 0. \end{cases}$$

Use the integer clock of [3] on the complete finite state

$$\Theta_c(X) = -\sigma(X) \sum_{q \in B} \chi(q) f(q_x + q_y - c) + G(p_x + p_y - c, \epsilon(h)). \tag{31}$$

Since $f(s)$ is even, Θ_c is an exactly computable integer.

Lemma A.1 (Integer clock, Ke [3])**.** *For every finite-support state and integer centre c, with e, e' the before and after values of ϵ and $s = p_x + p_y - c$,*

$$\Theta_c(FX) - \Theta_c(X) = 1 + 4 f(s) \mathbf{1}_{\{e' \neq e,\ es < 0\}} \ \geq 1. \tag{32}$$

Proof. Write $\delta = +1$ on R and -1 on L. Directly checking the two colours at four headings gives $\delta \sigma \chi(p) = ee'$, and the outgoing step gives $s' = s + e'$. Thus the black-field increment is $-ee' f(s)$. Let $G_\pm(s) = G(s, \pm 1)$. The elementary recurrences are

$$G_+(s+1) - G_+(s) = f(s) + 1, \qquad G_-(s-1) - G_-(s) = f(s) + 1,$$
$$G_-(s) - G_+(s) = -2 \operatorname{sgn}(s) f(s).$$

If $e' = e$ the total gain is one. For $+1 \to -1$ it is $1 + 2f(s) + G_-(s) - G_+(s)$, equal to one for $s \geq 0$ and to $1 + 4f(s)$ for $s < 0$. The other reversal gives the reflected cases. This includes $s = 0$. □

A.2 Diagonal wake budgets and crossing refinement

Rotate a diagonal drift to $d = (m, -m)$, $m > 0$, and use the transformed heading. Let $t = x + y$, let o_t count odd classes at level t, and define

$$A_t = -\sigma(-1)^t o_t, \qquad K_c(O) = \sum_t A_t(3^{|t-c|} - 1). \tag{33}$$

Theorem A.2 (Wake-determined diagonal period)**.** *Every realization of this wake obeys $P \le K_c(O)$ for every integer c.*

Proof. Over one period the head term of (31) cancels, since both t and the heading return to their initial values. The signed cell toggles give $-\sigma \sum_i \chi(q_i) f(t_i - c)\delta_i$. The weight is constant in each full d-class. Lemma 2.1 makes its net δ count one on odd classes and zero on even classes. The exact period increment is therefore $K_c(O)$. Summing (32) proves the bound. The unchanged part of the full field cancels from the period increment. □

Here is an independent geometric derivation, which also supplies a refinement. The finite projected period is a closed walk with increments ± 1 in t. Let f_t count upward crossings of the cut $(t, t+1)$; it also counts downward crossings. Let m_t count visits to t, and u_t count passages entering and leaving in the increasing direction. Flow conservation equates this with the number of decreasing-through passages. The fixed arrival axis and the two turn choices give

$$m_t = f_{t-1} + f_t = A_t + 4u_t, \qquad 0 \le u_t \le \min(f_{t-1}, f_t). \tag{34}$$

For example, at vertical-arrival levels, right turns are through passages and left turns reverse the projection, giving $\#\mathrm{R} - \#\mathrm{L} = 4u_t - m_t$; horizontal arrival reverses that sign. Substituting $u_t \le f_{t-1}$ into $f_t = A_t + 4u_t - f_{t-1}$ gives the first inequality below; $u_t \le f_t$ gives the second:

$$f_t \le 3f_{t-1} + A_t, \qquad f_{t-1} \le 3f_t + A_t. \tag{35}$$

Let ℓ, h be the extreme wake levels. Outside them $A_t = 0$; starting past the finite support of the projected walk and applying (35) inward forces $f_{\ell-1} = f_h = 0$. Thus the periodic trace lies between the extreme wake levels. An empty wake forces every $f_t = 0$, impossible for a nonempty period.

Starting from these two zero boundary values, set

$$L_{\ell-1} = 0, \quad L_t = 3L_{t-1} + A_t, \qquad R_h = 0, \quad R_t = 3R_{t+1} + A_{t+1}, \quad C_t = \min(L_t, R_t).$$

Then $0 \le f_t \le C_t$ for $\ell \le t < h$. If a C_t is negative, the wake is impossible; otherwise

$$P = 2\sum_t f_t \le 2\sum_t C_t. \tag{36}$$

Moreover $d_x = \sigma \sum_t (-1)^t f_t$: on a cut with $\sigma(-1)^t = 1$, the upward horizontal moves contribute $+f_t$ to x; on the other cuts the downward horizontal moves contribute $-f_t$. If C_+ and C_- sum the corresponding positive and negative cut bounds, then also

$$P \le 4C_+ - 2d_x, \qquad P \le 4C_- + 2d_x. \tag{37}$$

Corollary A.3 (Width and growth bound)**.** *Writing $W = h - \ell$, a diagonal realization has even W and*

$$P \le g(3^{W/2} - 1). \tag{38}$$

Proof. At a transverse extreme every passage reverses the projection. The fixed arrival axis makes all letters in an extremal class equal. Lemma 2.1 therefore forces exactly one R per such class. Both extremes have the same arrival axis, so W is even; this is the usual extremal rigidity argument. Take $c = (\ell + h)/2$ in Theorem A.2, discard negative coefficients and use $\sum_t \max(A_t, 0) \le g$ to obtain (38). □

For the normalized standard wake, $\sigma = -1$ and the coefficients at $t = -10, \dots, 0$ are

$$A = (2, -1, 1, -1, 1, -1, 1, 0, 1, -1, 2).$$

The cut bounds are $C = (2, 5, 16, 47, 142, 145, 48, 16, 5, 2)$, with sum 428. Thus (36) gives $P \le 856$, as does (33) with centre $c = -5$. The chronological constraints in Section 4 strengthen this budget to the cutoff used in the complete classification.

A.3 General non-axis drift by reciprocal moments

For a general drift, use a transverse linear coordinate constant along it. An exponential weight and its reciprocal give complementary period budgets; Cauchy–Schwarz then bounds the number of steps independently of their individual weights.

Rotate the drift to $(a_0, -b_0)$, $a_0, b_0 > 0$. Put $k = \gcd(a_0, b_0)$, $\eta = (b_0/k, a_0/k)$, and choose rational $r > 1$ such that, with $a = r^{\eta_x}$ and $b = r^{\eta_y}$,

$$\kappa = 1 - \frac{2}{a+1} - \frac{2}{b+1} > 0.$$

The choice $r = 4$ always works. Define $w(q) = r^{\eta \cdot q}$, and

$$(\gamma_N, \gamma_E, \gamma_S, \gamma_W) = \left(\frac{2}{b+1}, \frac{2}{a+1}, -\frac{2b}{b+1}, -\frac{2a}{a+1} \right).$$

The one-sided potentials are

$$\Psi_+(X) = -\sigma \sum_{q \in B} \chi(q) w(q) + \gamma_h w(p), \qquad \Psi_-(X) = -\sigma \sum_{q \in B} \frac{\chi(q)}{w(q)} + \frac{\gamma_{h+2}}{w(p)}.$$

Substitution of the current-cell flip gives $\Delta\Psi_+ = \kappa w(p)$ in six heading/colour cases and $(\kappa + 4)w(p)$ in S/L and W/R. The half-turn formula gives $\Delta\Psi_- \ge \kappa / w(p)$. These local identities are the one-sided exponential clock construction, included to make the new period argument self-contained.

Define the two finite wake moments, using any representatives of O,

$$A_+ = -\sigma \sum_{q \in O} \chi(q) r^{\eta \cdot q}, \qquad A_- = -\sigma \sum_{q \in O} \chi(q) r^{-\eta \cdot q}.$$

They are independent of the representatives since $\eta \cdot d = 0$ and $d_x + d_y$ is even. The head terms cancel over a period, and Lemma 2.1 gives exact increments A_+ and A_-. Thus both are positive and $\kappa \sum_i w(q_i) \le A_+$, $\kappa \sum_i w(q_i)^{-1} \le A_-$. Cauchy–Schwarz yields:

Theorem A.4 (Reciprocal wake bound)**.** *Every realization with the specified non-axis drift and wake satisfies*

$$P^2 \le \frac{A_+ A_-}{\kappa^2}. \tag{39}$$

All quantities are rational. The integer square root of the floor of the right side is a computable cutoff. Changing the transverse centre scales A_+ and A_- reciprocally, leaving the product unchanged. The non-axis hypothesis ensures $\kappa > 0$.

A.4 All-period synthesis and detour obstruction

Theorem A.5 (Period-unknown inverse synthesis)**.** *Given a non-axis integer drift, an initial heading and a finite odd-class wake, existence of a finite-support periodic realization is decidable. A positive answer can produce its word and an explicit finite seed.*

Proof. First reject odd $d_x + d_y$, growth $|O|$ not divisible by four, or a nonpositive moment A_+ or A_-. Otherwise compute Theorem A.4's cutoff, additionally using Theorem A.2 and (36)–(37) in the diagonal case. A negative crossing bound also rejects the input. Enumerate every nonempty word up to the least derived bound. Check its heading and drift, apply Lemma 2.1, and require the exact odd-class set. A passing word constructs a finite seed; complete finite exhaustion gives a negative answer. Completeness follows from the period bound. □

The all-future verifier of Section 10 independently checks the constructed finite seed. For cutoff B, enumeration visits at most $2^{B+1} - 2$ nonempty binary words; the period bound guarantees termination.

Corollary A.6 (No wake-preserving closed-pose insertion)**.** *Suppose realizable words u, v have the same non-axis drift, and v is obtained by inserting nonempty detours into u, each returning to the same position and heading while leaving the original steps unchanged. Then*

$$A_+(O_v) > A_+(O_u), \qquad A_-(O_v) > A_-(O_u).$$

Thus their signed transverse wake cannot be identical. For diagonal drift, $K_c(O_v) - K_c(O_u) \geq |v| - |u| > 0$ for every integer c.

Proof. Every retained original step has the same position, heading and letter; hence it has the same local clock increment. Every inserted step has strictly positive increment in both one-sided clocks, and at least one in the integer clock. Sum over the words and use the exact wake-increment equalities. □

Thus any such realizable detour requires an additional wake budget.

B Mixed-wake interleaving at arbitrary width

The thin-wake classification uses a particular growth and width. A separate parity identity constrains arbitrary finite odd-class wakes at the same full drift $(2, -2)$, without a period or width cutoff. It tests the transverse order of the classes, in addition to their signed residue counts.

Translate a vertical-arrival phase to the origin. Then even levels $t = x + y$ have vertical arrivals, denoted Q, and odd levels have horizontal arrivals, denoted C. Lanes are $r = x \bmod 2$. Let O be the odd classes, $g = |O|$, and define

$$V = |O \cap Q|, \qquad J(O) = \#\{(q, c) \in (O \cap Q) \times (O \cap C) : t(q) < t(c)\}.$$

Distinct classes on the same layer are counted separately.

Theorem B.1 (Mixed-wake interleaving)**.** *Every finite-support periodic realization of full drift $(2, -2)$ satisfies*

$$V \equiv 0 \pmod 2, \qquad J(O) + V/2 \equiv 1 \pmod 2.$$

The condition can be evaluated in $O(g \log g)$ integer operations by sorting only the occupied odd classes, irrespective of transverse gap sizes.

B.1 Signed residues and the rank factorization

We use the stable-class and signed-residue arguments of Jillhewar [2, Section 7, "Signed mod-four wake-residue identity", and the proof of "Even-diagonal growth four is impossible", equation labelled `eq:diagK`]. The supplemental `proof/periodic-growth-closure.md` supplies an attributed derivation, including an explicit mod-four potential. For a function $f : \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, choose $S(0) = 0$ and $S(x+1) - S(x) = f(x) + f(x+1)$. With $\chi = (-1)^{x+y}$, put

$$F_N = -\chi S(x), \qquad F_E = \chi(S(x) - f(x)),$$
$$F_S = -\chi(S(x) + 2f(x)), \quad F_W = \chi(S(x) + f(x)).$$

The complete-state quantity $\sum_{(x,y)\in B} \chi(x,y)f(x) + F_h(p)$ is conserved modulo four, as direct substitution in the eight heading/colour cases verifies. For a residue indicator modulo a nonzero horizontal displacement a, its monodromy gives

$$\sum_{q\in O,\ q_x = r \bmod |a|} \chi(q) \equiv 2 \pmod 4.$$

Quarter-turning gives the vertical counterpart. In particular, with $q_r = |O \cap Q \text{ in lane } r|$ and $c_r = |O \cap C \text{ in lane } r|$,

$$q_0 - c_0 \equiv q_1 - c_1 \equiv q_0 - c_1 \equiv q_1 - c_0 \equiv 2 \pmod 4.$$

Thus V is even and $V/2 \equiv q_0 \pmod 2$.

The phase successor acts on the P period events. Order those events first by translation class and then by stable rank within each class. The actual map sends phase i to phase $i+1 \pmod P$, so it is one P-cycle. To factor it in the class order, match incoming rank to outgoing rank at each local turn, sort outgoing events by their port bits, transfer each directed edge block in its preserved order, and undo the incoming-port sorting at the target. Multiplying the signs of these permutations gives the relation used below. The same factorization applies to the general drift in Appendix C.

For completeness we evaluate the local signs. At each class, order visits by stabilized rank k, and record positive incoming and outgoing port bits α_k, β_k: the positive ports are N/E at Q, E/N at C. For a binary string with m positions and A ones,

$$\operatorname{inv}(\alpha) \equiv A(m-1) + \sum_k k\alpha_k + \binom{A}{2} \pmod 2.$$

Apply this to both port strings. Directed-edge binomial terms cancel between source and target. If D_v is the odd-class bit and $p_v = \sum_k(\alpha_k + \beta_k) \bmod 2$, the remaining local correction is $p_v(D_v + 1)$ at Q, and $D_v(p_v + 1)$ at C. Denote its sum by $\mathcal{C}$, and the edge-block permutation sign by $\mathcal{E}$. Stable order within each directed edge block is preserved, since its successor time changes by the fixed amount $1 - P$ voltage. The phase successor is one even-length cycle. Its factorization consequently gives

$$K := \sum_{\text{L events}} \text{rank} \equiv 1 + \mathcal{E} + \mathcal{C} \pmod 2.$$

For an actual stable class, L has odd rank. Also $4 \mid P$ and $4 \mid g$, so $K \equiv (P - g)/2 \equiv 0 \pmod 2$. Hence $\mathcal{E} + \mathcal{C} = 1$. This factorization is inherited; the following evaluation for general mixed wakes is the extension.

B.2 A Boolean identity with a telescoping boundary term

All arithmetic in this subsection is over $\mathbb{F}_2$. Write $Q_{j,r} = D_{(2j,r)}$, $C_{j,r} = D_{(2j+1,r)}$. Let e, w count E/W outgoing edges at Q modulo two, and n, s count N/S outgoing edges at C. Local balance gives, as two-vector identities,

$$w_j = e_j + Q_j, \quad s_j = n_j + C_j, \quad n_j = n_{j-1} + Q_j + C_j, \quad e_j = e_{j-1} + Q_j + \mathrm{swap}(C_{j-1}).$$

All counts vanish below and above the finite range of one period. We prove the closed identity

$$\mathcal{E} + \mathcal{C} = J(O) + \binom{g}{2} + q_0. \tag{*}$$

Encode a layer's outgoing parities in the order negative lane zero, positive lane zero, negative lane one, positive lane one. Its internal inversion contributions are

$$S_Q(b) = b_0b_2 + b_1b_2 + b_1b_3, \qquad S_C(b) = b_1b_2.$$

The adjacent-layer term is $A(a, b) = (a_1 + a_3)(b_0 + b_2)$. At separation two the terms are

$$B_Q(a, b) = a_1(b_0 + b_2) + a_3b_2, \qquad B_C(a, b) = a_1b_0 + a_3(b_0 + b_2).$$

Larger separations give no inversions. These expressions follow by ordering the directed targets: Q edges swap lane and C edges preserve it.

Before a new pair of layers, use state (e_0, e_1, c_0, c_1, z), where z is the accumulated Q population parity. The previous N parities are $n_0 = e_1 + z + c_0$, $n_1 = e_0 + z + c_1$. For new bits (Q_0, Q_1, C_0, C_1), the next state is $(e'_0, e'_1, C_0, C_1, z')$, where

$$e'_0 = e_0 + Q_0 + c_1, \quad e'_1 = e_1 + Q_1 + c_0, \quad n'_r = n_r + Q_r + C_r, \quad z' = z + Q_0 + Q_1.$$

Set

$$\begin{aligned} a &= (0, e_0, 0, e_1), & b &= (n_0 + c_0, n_0, n_1 + c_1, n_1),\\ u &= (e'_0 + Q_0, e'_0, e'_1 + Q_1, e'_1), & v &= (n'_0 + C_0, n'_0, n'_1 + C_1, n'_1). \end{aligned}$$

Then

$$\begin{aligned} \Delta\mathcal{E} &= S_Q(u) + S_C(v) + A(b, u) + B_Q(a, u) + A(u, v) + B_C(b, v),\\ \Delta\mathcal{C} &= (n_0 + e'_0)(1 + Q_0) + (n_1 + e'_1)(1 + Q_1)\\ &\quad + C_0(1 + e'_1 + n'_0) + C_1(1 + e'_0 + n'_1),\\ \Delta J &= z'(C_0 + C_1). \end{aligned}$$

For $d = (Q_0, Q_1, C_0, C_1)$, the population correction is

$$\Delta\binom{g}{2} = (n_0 + n_1)\sum_i d_i + \sum_{i<j} d_id_j.$$

The explicit boundary polynomial

$$H = e_0 + e_1 + z + e_0c_1 + e_1c_0 + e_1c_1 + e_1z + c_0c_1 + c_0z$$

satisfies

$$H(\text{old}) + H(\text{new}) = \Delta\mathcal{E} + \Delta\mathcal{C} + \Delta J + \Delta\binom{g}{2} + Q_0.$$

Substitution with $x^2 = x$ proves the identity in nine Boolean variables; the checker both expands the polynomial and tests all 512 assignments. Summing over any finite number of layer pairs telescopes H. Empty boundary padding makes it zero at both ends, giving (*). Since $4 \mid g$ and $q_0 \equiv V/2$, comparison with $\mathcal{E} + \mathcal{C} = 1$ proves Theorem B.1.

B.3 Growth-eight wakes with two full extreme layers

Suppose $g = 8$ and both residue classes at each transverse extreme are odd. Extreme visits are singleton right turns, hence on C layers. They supply four C classes. Thus V is even and at most four. If $V = 0$, the new parity is zero. If $V = 4$, all C classes are at the extremes; each Q lies below exactly two of them, so $J = 8$ and $J + V/2$ is again even. Therefore $V = 2$, with lane populations $(q_0, q_1) = (1, 1)$, $(c_0, c_1) = (3, 3)$. The x/y residue matrix is $\left[\begin{smallmatrix} 1 & 3 \\ 3 & 1 \end{smallmatrix}\right]$.

If the four internal odd classes have distinct levels, their increasing Q/C sequence must have an even number of Q-before-C pairs. The alternating orders QCQC and CQCQ are excluded for arbitrary gaps and lane choices. QQCC, QCCQ, CQQC and CCQQ pass this particular test; they are candidates requiring the full realization criterion. This consequence does not require a transverse-width bound.

The exact controls include 18,895 geometric heading-restoring words, 65,536 finite layer subsets, all 104 standard phases, and an independent edge-permutation computation. A separate bounded screen at widths 10,12,14 excludes respectively 3,058, 11,374 and 33,320 assignments passing the signed residues and checker-mixing filters. Those finite tables illustrate the screen; the telescoping identity proves it at every width. The complete periodic-growth supplement also retains attributed small-drift exclusions and the regenerated unit-diagonal certificate, keeping their ownership and logical dependencies explicit.

C Quadratic wake tests for non-axis drift

The rule is white/right or black/left, followed by toggling the current site and moving one edge. The black field is a complete finite subset of $\mathbb{Z}^2$. Suppose a nonempty turn word of length P repeats forever, restores the heading, and has nonzero full displacement d. Finite preludes and genuinely unread debris are allowed. We do not replace d by its primitive direction.

The stable-order and port-factorization method of [2] lets us eliminate the unknown event graph and test the wake directly. Assume $d = (a, b)$, $ab \neq 0$. A half-turn makes $a > 0$; heading restoration implies $a + b$ even. Let O consist of full d-classes with odd phase multiplicity and write $g = |O|$. Canonical representatives are

$$\pi_d(x, y) = (r, y - b\lfloor x/a \rfloor), \qquad r = x \bmod a, \quad 0 \le r < a.$$

Checker parity $c(r, y) = (r + y) \bmod 2$ is well-defined on the quotient. Initially use $Q = \{c = 0\}$ for vertical-arrival vertices and $C = \{c = 1\}$ for horizontal arrivals. Lemma C.4 removes dependence on this convention.

C.1 The graph reconstructed from a wake

Suppose that every horizontal and vertical quotient component contains an even number of O vertices. On each vertical line $r = x \bmod a$, sort the canonical y coordinates and pair consecutive endpoints by finite intervals. Horizontal components have representative row $Y = y \bmod |b|$. Express a point there using

$$k = (y - Y)/b, \qquad X = x - ka,$$

then sort and pair the X coordinates. Let $G_d(O)$ be the union of these horizontal and vertical interval edges. Orient horizontal edges from Q to C and vertical edges from C to Q.

Order vertices by (y, r) and ports negative before positive; positive means movement E or N. An edge has source and target keys $(\text{vertex}, \text{port})$. Let $E_d(O)$ count, modulo two, edge pairs whose

source and target orders disagree. Let $N_+(O)$ count positive retained edges, and put $C(O) = |O \cap C|$. Define

$$\mathfrak{q}_d(O) = E_d(O) + N_+(O) + C(O) \pmod 2, \tag{40}$$

$$\Omega_d(O) = \sum_{(r,y)\in O} \lfloor y/|b| \rfloor \pmod 2. \tag{41}$$

The finite graph may be enormous, but Section C.4 does not construct its individual edges.

Theorem C.1 (Wake obstruction)**.** *Every finite-support periodic realization with non-axis full drift satisfies*

$$\boxed{\Omega_d(O) = 1, \qquad \mathfrak{q}_d(O) = 1 + a \pmod 2.}$$

There is no restriction on period, transverse width, growth, retreat, seed size, or wake shape. The characters are independent of representatives, translation, phase origin and checker/heading-axis convention; no heading is a screen input.

Theorem C.2 (Quadratic structure)**.** *On the $\mathbb{F}_2$ vector space of finite indicators with even sums in every horizontal and vertical quotient component, $\mathfrak{q}_d$ is a quadratic form. In particular $\mathfrak{q}_d(A + B) + \mathfrak{q}_d(A) + \mathfrak{q}_d(B)$ is bilinear. Addition is symmetric difference of indicators, not superposition of physical trajectories.*

Theorem C.3 (Compressed complexity)**.** *Given d and explicit representatives of O in binary, signed-residue prefilters followed by Theorem C.1 are computable in polynomial bit time. After the density filter, at most $R = g(a+1) \le g(g/2+1)$ arithmetic edge runs suffice, and the evaluator uses $O(g^4 \log(g+1))$ integer arithmetic operations on $O(B + \log(g+1))$-bit integers, where B bounds coordinate and drift bit lengths. Passing means only `NOT_EXCLUDED`, not realizability.*

C.2 Boundary reconstruction and winding

In a full d-class write phase positions $q_i = b_v + k_i d$. A sufficiently distant cell $b_v + nd$ is initially white and eventually has all the phase visits, at times $nP + i - Pk_i$. The necessary stable-class condition of [2] therefore orders visits by $\tau_i = i - Pk_i$ and gives letters $R, L, R, L, \ldots$. The signed excess in a class is its odd indicator $D_v \in \{0, 1\}$. Thus $g = \sum D_v$, and heading restoration gives $4 \mid g$.

The checker/heading-axis invariant assigns incoming axes alternately to the two checker colours. Every visit uses one horizontal and one vertical edge. Modulo two, the horizontal incident-edge boundary and the vertical incident-edge boundary are consequently both D.

Because $b \neq 0$, the horizontal quotient graph is a union of infinite paths: no nonzero horizontal vector is in $\mathbb{Z}d$. The analogous fact holds vertically because $a \neq 0$. On an infinite path, a finite edge set with prescribed boundary is unique: its indicator is the cumulative endpoint parity, hence exactly the intervals pairing consecutive endpoints. A nonempty finite boundary-free edge set on a line cannot exist. Therefore the actual edge-count parities are precisely $G_d(O)$, whatever their unknown even parts may be.

This argument explains the domain restriction. For axis drift, one direction has quotient cycles, and a boundary-free cycle can be added to parity data. See Section 11 for the resulting axis-scope boundary.

The actual period has signed seam crossing number one across the cut from x residue $a - 1$ to 0. For a horizontal interval from X_L to X_R, its seam crossing parity is $\lfloor X_R/a \rfloor - \lfloor X_L/a \rfloor$. Summing over paired intervals equals the sum of these floors over all endpoints modulo two. A canonical

endpoint (r, y) has $X = r - ak$, with $k = (y - (y \bmod |b|))/b$, so $\lfloor X/a \rfloor = -k$ has parity $\lfloor y/|b| \rfloor$. This proves (41) and $\Omega_d = 1$. On arbitrary representatives the same weight is $\lfloor x/a \rfloor + \lfloor y/|b| \rfloor \pmod 2$; adding d changes it by the even integer $1 + \mathrm{sign}(b)$.

C.3 The one-tour sign and its wake reduction

At a vertex order its m events by stable rank $k = 0, \dots, m-1$. Let α_k, β_k be positive incoming and outgoing port bits, with totals A, B, and put $p_v = A + B \pmod 2$. At Q, a left turn means unequal bits; at C, it means equal bits. For any binary list,

$$\mathrm{inv}(\alpha) = A(m-1) + \sum_k k\alpha_k + \binom{A}{2} \pmod 2. \tag{42}$$

Indeed, each one at position k sees $m-1-k$ later positions, from which its later ones are subtracted. Summing the latter gives $\binom{A}{2}$.

Put $K = \sum_{L \text{ events}} \text{stable rank} \pmod 2$. Equation (42) applied to both port lists leaves local corrections $p_v(m-1)$ at Q and $p_v(m-1) + \binom{m}{2}$ at C, besides port inversions and positive-edge binomials. At Q, $m \equiv D_v \pmod 2$. At C, counting unequal bits gives

$$D_v = 2(A+B) - 4\sum_k \alpha_k\beta_k - m, \qquad m \equiv 2p_v - D_v \pmod 4.$$

For $D_v = 0, 1$ the two corrections are respectively $p_v(D_v+1)$ and $D_v(p_v+1)$. The positive-edge binomials cancel between source and target. If I is the sum of all incoming/outgoing port-list inversion parities,

$$K = I + L, \qquad L = \sum_Q p_v(D_v+1) + \sum_C D_v(p_v+1). \tag{43}$$

Along a fixed directed quotient edge of voltage ν, target stable time is source stable time plus $1 - P\nu$. The same constant applies to every occurrence, including the cyclic seam. Occurrences therefore match first-to-first. Sorting outgoing ports, following edge blocks and unsorting incoming ports factors the phase successor. This successor is *one* even-length P-cycle, of odd sign; hence

$$I + E = 1, \qquad K = 1 + E + L. \tag{44}$$

A block reversal of sizes u, v contributes $uv \pmod 2$. Thus E depends only on edge-count parities and equals $E_d(O)$. This is the general version of the attributed special-drift factorization, not a finite-period extrapolation.

Lemma C.4 (Cancellation and conventions)**.** *For $G_d(O)$ one has*

$$\sum_{v \in O} p_v = 0, \qquad L = N_+(O) + C(O) \pmod 2.$$

The character is unchanged by the total vertex order or by exchanging the checker/heading-axis convention.

Proof. The positive incidence of a paired horizontal interval at its two endpoints has parity $1 + c(\text{left}) + c(\text{right})$. A vertical interval has the same formula. There are $g/2$ intervals of each kind; each endpoint checker bit is counted twice. The total is $g = 0 \pmod 2$. Also $\sum_Q p_v = N_+$, since every positive edge touches exactly one Q vertex. Expanding (43) proves the stated expression for L.

Swapping vertices v, w in the order changes E by $\operatorname{out}(v)\operatorname{out}(w) + \operatorname{in}(v)\operatorname{in}(w) = 2D_vD_w = 0 \pmod 2$. Exchanging checker conventions reverses all edges and complements the ports. Inversion sign is invariant under permutation inversion; the extra port-block reversals contribute $\sum_v (1 + D_v)p_v = 0$, by endpoint cancellation and $\sum_v p_v = 0$. The edge set has even size, because the sum of outgoing degree parities is g. Thus replacing N_+ by $|G| - N_+$ does not change its parity; neither does replacing C by $g - C$. □

Translations merely relabel vertices and possibly exchange conventions, so $\mathfrak{q}_d$ is translation and phase invariant. Moving a seam does not change winding, because every fibre population is even.

For an actual stable class word, every L has odd rank. Hence $K = \#L = (P - g)/2 \pmod 2$. There are $P/2$ horizontal moves with net displacement a, so $P/2 \equiv a \pmod 2$; since $4 \mid g$, $K \equiv a$. Equations (40), (43), (44) prove Theorem C.1.

The reconstruction map $D \mapsto G_d(D)$ is linear by uniqueness of a finite boundary antiderivative. The inversion parity is quadratic in edge indicators, and N_+, C are linear. This proves Theorem C.2. For $(2, -2)$, the previous evaluation in Appendix B is $\mathfrak{q}_d = J + V/2$ under signed residues, where J counts Q-before-C odd-class pairs in transverse order and $V = |O_Q|$. Thus the earlier formula $J + V/2 = 1$ is recovered.

C.4 Compressed evaluation

The signed residue theorem [2, Section 7, "Signed mod-four wake-residue identity"] gives, in every x residue modulo a and every y residue modulo $|b|$, signed odd-class sum 2 (mod 4), with sign $(-1)^{x+y}$. In particular $g \geq 2\max(a, |b|)$. Check positivity and $4 \mid g$, then density *before* allocating residue-sized arrays; rejected inputs need no graph construction. Both drift magnitudes are then at most $g/2$. Signed sums also enforce even fibre populations.

An arithmetic edge run has n edges with source and target keys

$$(s_y + hi, s_x, p), \quad (t_y + hi, t_x, p), \qquad 0 \leq i < n, \quad h > 0. \tag{45}$$

A vertical interval splits by bottom-site parity into at most two step-two runs. For a horizontal interval on representative row Y, split its left positions $X = r + ak$ by r, and, for odd a, by k mod 2. Canonical heights change by $-b$ per k. Reverse the index if necessary to obtain positive step $|b|$ or $2|b|$ and constant port bits. There are at most $2a$ runs per horizontal interval, so $R \leq g(a + 1)$. All runs have disjoint edges.

A run has no internal inversions. Between runs A, B, set

$$z = s_A + h_A i - s_B - h_B j, \qquad \delta = (t_A - s_A) - (t_B - s_B).$$

Let $s = 0$ if the source minor key (x_A, p_A) precedes (x_B, p_B), otherwise $s = -1$. Define $u = -\delta$ plus 0 or -1 by the analogous target comparison. The edge pair is inverted exactly when

$$\min(s, u) < z \leq \max(s, u). \tag{46}$$

Two cumulative pair counts suffice. For $ui - vj \leq T$, $0 \leq i < n$, $0 \leq j < m$, put

$$\ell = \operatorname{clip}(\lceil -T/v \rceil, 0, m), \qquad h = \operatorname{clip}(\lceil (un - T)/v \rceil, 0, m).$$

The exact number of pairs is

$$(m - h)n + (h - \ell) + \sum_{j=0}^{h-\ell-1} \left\lfloor \frac{vj + T + v\ell}{u} \right\rfloor. \tag{47}$$

There are no eligible i below ℓ, all n are eligible above h, and the middle contributes $\lfloor (T+vj)/u \rfloor + 1$.

The floor sum $S(n, m, a, b) = \sum_{i=0}^{n-1} \lfloor (ai + b)/m \rfloor$ uses Euclidean lattice transposition. Remove quotient parts of a, b, contributing $\lfloor a/m \rfloor n(n-1)/2 + \lfloor b/m \rfloor n$. For $0 \le a, b < m$ the remainder is zero if $an + b < m$; otherwise, with $n' = \lfloor (an + b)/m \rfloor$ and $b' = (an + b) \bmod m$,

$$S(n, m, a, b) = S(n', a, m, b'). \tag{48}$$

At positive height j there are $n - \lceil (mj - b)/a \rceil$ eligible indices. Reindex heights from n' downwards, using $mn' - b = an - b'$, to obtain the right side of (48). Division first handles negative offsets exactly. Euclidean reduction needs logarithmically many stages.

All run-pair counts have polynomial bit length even for runs of 10^{100} edges. Summing positive run lengths gives N_+; C and Ω_d use only the endpoints. The run bound, R^2 run pairs, and (47)–(48) prove Theorem C.3. A passing wake proceeds to the chronology-exact inverse synthesis of Appendix A.

C.5 An unbounded growth-eight exclusion

Let $d = (3, 1)$ and

$$O_* = \{(0,1), (0,3), (1,-2), (1,-1), \\ (1,0), (1,2), (2,0), (2,2)\}.$$

For $n \ge 1$, put $\lambda = 12n + 1$ and let O_n be the classes of λq, $q \in O_*$. They are eight distinct classes with canonical representatives

$$(r, (12y - 4r)n + y), \qquad (r, y) \in O_*.$$

Their signed x-fibre sums are $(-2, -2, 2)$ and their sole y-fibre sum is -2. They have growth eight, pass density, and have winding $\sum \text{canonical } y = 28n + 5 \equiv 1 \pmod 2$.

Corollary C.5. *No finite-support periodic highway of drift $(3, 1)$ has wake O_n, for any integer $n \ge 1$ and any period. Their algebraic transverse span, measured by $T(x, y) = 3y - x$, is $16(12n + 1)$ and is unbounded.*

Proof. The affine runs in Appendix C.7 give

$$|G_d(O_n)| = 192n + 16, \quad N_+(O_n) = 96n + 9, \quad \#\text{inversions} = 400n + 19, \quad C(O_n) = 5.$$

Thus $\mathfrak{q}_d(O_n) = 496n + 33 \equiv 1$, whereas Theorem C.1 requires $1 + a \equiv 0$. For completeness, the calculation is an all-parameter integer identity, not an interpolation. If $L < R$ are horizontal representative-row endpoints before dilation, the index bounds at residue r are

$$k_{\min} = 4Ln + \lceil (L - r)/3 \rceil, \qquad k_{\max} = 4Rn + \lfloor (R - 1 - r)/3 \rfloor.$$

Splitting parity gives 32 runs of step two and positive affine lengths on $n \ge 1$. For a parity-compatible inversion difference z, set $k = (z - s_A + s_B)/2$. Its count is exactly

$$\max\{0, \min(n_B, n_A - k) - \max(0, -k)\}. \tag{49}$$

Every minimum/maximum branch is verified uniformly: an affine expression $An + B$ is nonnegative for every $n \ge 1$ exactly when $A \ge 0$ and $A + B \ge 0$. Summing the 288 parity-compatible terms among 496 run pairs gives $400n + 19$. The supplement implements this symbolic arithmetic and fails rather than extrapolating if a comparison changes sign on the parameter domain. □

The unscaled O_* also fails: its counts are 16 retained edges, 9 positive edges, 23 inversions and $C = 5$. For contrast,

$$\{(0,1),(0,5),(1,-4),(1,-2),(1,5),(1,6),(2,-4),(2,4)\}$$

has the same signed fibre populations and winding, but $\mathfrak{q}_d = 0$. It satisfies this necessary parity condition. Thus the character strengthens the specified signed-residue, density and winding conditions. Physical realization additionally requires chronological compatibility of the visits.

C.6 Exact computational checks

The proof and complexity theorem do not depend on bounded experiments. Exact affine algebra establishes the entire family parameter range. Independent implementations check 10,000 floor sums and 10,000 pair-count instances against literal sums, 4,000 individual-edge reconstructions for eight drift types, vertex-order and checker-convention independence, and quadratic polarization. All 8,192 closed masks in an eight-layer/two-lane window reproduce in Appendix B.

An independent C++ vector-turn implementation enumerates all 1,398,100 binary words of even length at most 20 and checks the unspecialized port factorization on all 456,120 non-axis heading-restoring words. The 84 words with class balances in $\{0,1\}$ are additionally cross-checked against the wake-only formula. These geometric words are not asserted physically realizable. All 104 standard-highway phases, four rotations and powers one through four provide 1,664 positive controls using their *full* displacements. Representatives with large coordinates and $n = 10^{100}$ are checked without traversing their gaps; the latter still uses 32 runs. Exact reports and source manifests are in the supplement.

C.7 Affine edge-run certificate

All runs have step two and index $i = 0, \dots, \ell - 1$. A row gives length ℓ, source $(s_x, s_y + 2i)$, target $(t_x, s_y + \delta + 2i)$ and port p. These are edges of the parity graph, not time intervals of a proposed ant orbit. The row order is only an enumeration of runs; inversions use vertex order (y, r). Formula (49) on these rows yields the counts in Corollary C.5 for the whole domain $n \geq 1$.

Run	ℓ	s_x	s_y	t_x	δ	p
1	$8n+1$	0	$20n+2$	1	0	1
2	$8n+1$	1	$20n+3$	0	0	0
3	$8n$	2	$20n+4$	1	0	0
4	$8n+1$	1	$20n+3$	2	0	1
5	$8n$	2	$20n+4$	0	-1	1
6	$8n+1$	0	$20n+2$	2	1	0
7	$2n$	0	$12n+2$	1	0	1
8	$2n$	1	$12n+3$	0	0	0
9	$2n$	2	$12n+2$	1	0	0
10	$2n$	1	$12n+3$	2	0	1
11	$2n+1$	2	$12n+2$	0	-1	1
12	$2n$	0	$12n+2$	2	1	0
13	$2n$	0	$-8n$	1	0	1
14	$2n$	1	$-8n+1$	0	0	0
15	$2n+1$	2	$-8n$	1	0	0

Run	ℓ	s_x	s_y	t_x	δ	p
16	$2n$	1	$-8n+1$	2	0	1
17	$2n$	2	$-8n+2$	0	-1	1
18	$2n$	0	$-8n$	2	1	0
19	$6n+1$	0	$-28n-2$	1	0	1
20	$6n$	1	$-28n-1$	0	0	0
21	$6n$	2	$-28n$	1	0	0
22	$6n+1$	1	$-28n-1$	2	0	1
23	$6n$	2	$-28n$	0	-1	1
24	$6n+1$	0	$-28n-2$	2	1	0
25	$12n+1$	0	$12n+3$	0	-1	0
26	$12n+1$	0	$12n+1$	0	1	1
27	$6n+1$	1	$-28n-2$	1	1	1
28	$6n$	1	$-28n$	1	-1	0
29	$12n+1$	1	$-4n$	1	1	1
30	$12n+1$	1	$-4n+2$	1	-1	0
31	$12n+1$	2	$-8n+1$	2	-1	0
32	$12n+1$	2	$-8n+1$	2	1	1

D Analytic exclusion of the unit diagonal

The classical rule is: read white/right or black/left, turn, toggle the current cell, and move one edge on the entire square lattice. The initial black set is finite. Suppose a nonempty word u repeats forever, restores the heading, and has nonzero full displacement d. Set $P = |u|$ and $g = \#R - \#L$. The period need not be least; arbitrary unread finite debris is allowed.

Theorem D.1. *No such periodic highway has full displacement* $(1, -1)$ *or any quarter-turn rotation of this vector, regardless of period, width, growth, or finite seed.*

The full vector is essential: standard P104 has displacement $(2, -2)$ and is not excluded because its primitive direction is $(1, -1)$.

We prove the exclusion by forced layer words and a binary interleaving contradiction. This supplies an analytic alternative to the retained finite certificate.

D.1 Stable ranks and the singleton extreme

Write the phase positions as q_i, $0 \le i < P$. For $d = (1, -1)$, exactly one full translation class lies on each level $t = x + y$. Choose $b_t = (0, t)$ and write $q_i = b_t + a_i d$. At a sufficiently far cell $b_t + nd$, every phase is present, the initial colour was white, and the visit times are

$$nP + \tau_i, \qquad \tau_i = i - Pa_i.$$

Consequently each class, in increasing *stable rank* k, turns $R, L, R, L, \ldots$. This is the necessary part of the finite-support criterion of [2]. A fixed path of ℓ moves and displacement δd changes these stable times by $\ell - P\delta$.

The invariant σ is $(-1)^{x+y}$ for vertical incoming headings and its negative for horizontal incoming headings. Every step changes both factors. At a transverse extreme, the incoming axis and the inward-neighbor condition fix both directions of a visit. All turns of that class are identical, so

stable alternation forces one singleton R event. Choose the top event T as a sufficiently late phase origin and translate it to $(0, 0)$. It arrives east, leaves south, and gives $\sigma = -1$. The bottom arrives west and leaves north and must also have even transverse level. The width W is therefore positive and even. The new phase's complete field remains finite.

We will repeatedly use two direct consequences of chronology.

Forced matching. If a directed edge satisfies $b_s + e = b_t + \delta d$, it changes stable time by $1 - P\delta$. All source occurrences and all target occurrences have the same order and equal cardinality. They match first to first, not by an arbitrary permutation. At the period seam, the additional epoch cancels in $i - Pa_i$.

Consecutive-return rule. A fixed contiguous path's first return to its starting physical cell must increase that cell's stable rank by exactly one. Translate the path far enough along d for all its reads to be stable. Any intervening read would be another node of the same contiguous path; if none occurs, no stable rank can be skipped. If the endpoints were the same phase event, the path would instead form a closed component. Such a component cannot occur while obligatory deeper events remain: the successor on the P phase indices is one cycle, including when the word is a power of a shorter word.

D.2 The first two interior layers are forced

Build downwards. Let $a_k = 1$ indicate arrival from the already constructed upper side and $b_k = 1$ departure to that side. At even t the incoming headings for bits $1, 0$ are W,E, and at odd t they are S,N. Literal turning gives

$$b_k = a_k \oplus (k \bmod 2) \oplus (t \bmod 2). \tag{50}$$

If the word has length n, write $I = \sum a_k$, $J = \sum b_k$, $o = n \bmod 2$, and let e count arrival-ones at even ranks. Counting departures gives

$$n = \begin{cases} 2(I + J - 2e) + o, & t \text{ even}, \\ 2(J - I + 2e) - o, & t \text{ odd}. \end{cases} \tag{51}$$

The remaining lower cut counts are $n - J$ and $n - I$. They agree, starting with $I = J = 1$ from the top singleton.

Width zero is impossible because a move changes t. For $W = 2$, the extreme singletons force two middle visits, whereas (51) gives $2 = 4e - o$; since $n = 2$ implies $o = 0$, this is impossible. Thus $W \geq 4$, and layers $-1, -2, -3$ are strictly interior, with a deeper visited layer still required.

D.2.1 First layer

Denote rank k at level -1 by A_k. Here $I = J = 1$, so the only feasible arrival words are

$$100, \quad 1000, \quad 001, \quad 0010.$$

Indeed (51) gives $n = 4e - o$, and its single arrival-one lies at even rank zero or two. For either 001 or 0010, where the arrival-one has rank two, forced matching gives $A_0 \to T \to A_2$, with directions E,S. This travels by d in two updates, so $\tau(A_2) - \tau(A_0) = 2 - P < 0$, contrary to increasing rank. There are at least four distinct phase events, so $P \geq 4$. Only 100 and 1000 remain. Both have the fixed top path $A_2 \to T \to A_0$.

D.2.2 Second layer

Let B_k denote rank k at level -2. Let $p_0 < p_1 < \cdots$ list its arrival-one ranks and $q_0 < q_1 < \cdots$ its departure-one ranks. Each list has size $c = 2$ or 3, according as A has length three or four. Matching forces

$$B_{q_1} \xrightarrow{\mathrm{N}} A_2 \xrightarrow{\mathrm{E}} T \xrightarrow{\mathrm{S}} A_0 \xrightarrow{\mathrm{W}} B_{p_0}.$$

This is a first physical return, so the consecutive-return rule gives

$$p_0 = q_1 + 1. \tag{52}$$

At this even layer, every even rank belongs to both lists or neither, and every odd rank belongs to exactly one. Before p_0 there is no arrival-one. Thus every even rank there is in neither list, and every odd rank is a departure-one. Equation (52) forces

$$q_0 = 1, \qquad q_1 = 3, \qquad p_0 = 4.$$

The even rank four belongs to both lists, giving $q_2 = 4$. Therefore $c = 3$: A has word 1000. The departure list is exhausted; all later even ranks have arrival bit zero and all later odd ranks arrival bit one. The remaining arrivals must be at five and seven, and rank nine cannot occur. Hence B is exactly one of

$$00001101, \qquad 000011010. \tag{53}$$

No case enumeration is used in this forcing argument.

The fixed upper paths now include

$$B_1 \to A_1 \to B_5,$$
$$B_3 \to A_2 \to T \to A_0 \to B_4 \to A_3 \to B_7.$$

The B ranks departing downwards are $0, 2, 5, 6, 7$ and optionally eight; those arriving from below are $0, 1, 2, 3, 6$ and optionally eight. Only the first five of each list will be used, so the optional last event creates no further case.

D.3 A third-layer interleaving contradiction

Let C_k denote rank k at level -3 and reuse p_j, q_j for its arrival-one and departure-one ranks. There are at least five of each. Ordered matching forces the following paths:

$$C_{q_1} \to B_1 \to A_1 \to B_5 \to C_{p_2},$$
$$C_{q_3} \to B_3 \to A_2 \to T \to A_0 \to B_4 \to A_3 \to B_7 \to C_{p_4}.$$

Their compass words are respectively E,N,W,S and E,N,E,S,W,N,W,S. Both are first returns to the starting physical cell, with no other third-layer read between. Thus

$$p_2 = q_1 + 1, \qquad p_4 = q_3 + 1. \tag{54}$$

Two bridges, however, are forbidden. If $p_1 = q_1$, then

$$B_2 \to C_{p_1=q_1} \to B_1 \to A_1 \to B_5$$

has compass word S,E,N,W: its first physical return skips from rank two to five on B. If $p_4 = q_4$, then

$$B_4 \to A_3 \to B_7 \to C_{p_4=q_4} \to B_6$$

has compass word N,W,S,E: its first return skips from rank four to six. Both violate consecutivity, so

$$p_1 \neq q_1, \qquad p_4 \neq q_4. \tag{55}$$

Lemma D.2 (Binary interleaving)**.** *Let p and q be increasing lists of indices in $\{0, \dots, n-1\}$. Suppose every even index in this interval belongs to exactly one list, and every odd index to both or neither. If $p_2 = q_1 + 1$ and $p_4 = q_3 + 1$, then $p_1 = q_1$ or $p_4 = q_4$.*

Proof. If q_1 is odd, it also lies in p, and the next index is p_2, so $q_1 = p_1$. Otherwise q_1 is even and p_2 is odd, giving $p_2 = q_2$. If q_3 is even, p_4 is odd and gives $p_4 = q_4$. Otherwise q_3 is odd and is p_3. Avoiding both asserted equalities therefore gives $p_2 = q_2$ and $p_3 = q_3$, two distinct odd indices that are consecutive in both lists. No list member can lie strictly between them. But an even integer lies between distinct odd integers, and it must belong to exactly one list, a contradiction. □

At odd layer -3, equation (50) is precisely the list condition of Lemma D.2. Thus (54) contradicts (55).

Proof of Theorem D.1. Widths zero, odd widths, and width two were excluded directly. At every other width the first two interior words are forced by Section D.2, and their third-layer continuation is impossible by Section D.3. Quarter-turn rotation preserves the rule and finite support. No finite enumeration is a premise. □

D.4 Growth consequence and attribution

Jillhewar's results [2, Section 7, signed-residue theorem and the axis/even-diagonal growth-four theorems] give $g \in 4\mathbb{Z}_{>0}$ and $g \geq 2\max(|d_x|, |d_y|)$. For $g = 4$, both coordinates have absolute value at most two. Since heading restoration makes P even, $d_x + d_y$ is even. Up to quarter-turns, the only nonzero drift types are $(1, -1)$, $(2, 0)$ and $(2, -2)$. Theorem D.1 excludes the first; the cited all-period axis and even-diagonal growth-four theorems exclude the others. This recovers $g \geq 8$. The supplemental periodic-growth proof provides an attributed reconstruction of those external dependencies. Neither these earlier algebraic results nor the growth corollary is claimed as new here. The analytic unit-diagonal proof uses only the preceding stable-rank and interleaving arguments and does not rely on those other exclusions.

D.5 Independent computational checks

The supplement's `reproduce.py` reruns a complete-row Python enumerator, an independently written C++ vector-geometry enumerator, and witness audits. They agree on 4/124/12 756 local candidates and 2/2/0 survivors. A separate implementation validates all 12 880 rejected branches, 32 766 short binary words, 32 small count domains, eight compass port cases, two corrupt-witness negative controls, and 936 genuine P104 prefix controls retaining the full $(2, -2)$ quotient. An additional analytic audit verifies the displayed first return paths, the forced words (53), and the interleaving implication on every candidate in the finite enumeration. These checks corroborate the local identities; the symbolic forcing and contradiction above establish the all-period exclusion.

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